

Assignment_03

November 27, 2023

1 Authors

- Raphael Wäspi (18-918-938)
- Nicolas Huber (16-936-205)

```
[158]: import math
# Imports
import numpy as np
import matplotlib.pyplot as plt
import networkx as nx
import os
import glob
import re
import random
from statistics import mean

import scipy.special
from tqdm import tqdm
import shutil
from joblib import Parallel, delayed
from scipy.special import factorial
import powerlaw
from NEMtropy import DirectedGraph, UndirectedGraph
from NEMtropy import matrix_generator as mg
from NEMtropy.network_functions import build_adjacency_from_edgelist
```

```
[159]: # Define colors with reduced opacity from a colormap
COLOR_BLUE = 'dodgerblue'
COLOR_GREEN = 'palegreen'
COLOR_RED = 'lightcoral'
COLOR_YELLOW = 'gold'
COLOR_GREY = 'slategrey'
COLOR_BLACK = 'black'
COLOR_LIGHT_GRAY = 'lightgray'
COLOR_DARK_GRAY = 'darkgray'

# Define the main data directory
OUTPUT_PATH = "./output"
```

```

[160]: # Set Seed
SEED = 42

[161]: # Prepare output folder

# Delete the entire 'output' directory and its contents
if os.path.exists(OUTPUT_PATH):
    shutil.rmtree(OUTPUT_PATH)

# Create the 'output' directory
os.makedirs(OUTPUT_PATH)

# Define subdirectories
OUTPUT_PATH_ERGMS = os.path.join(OUTPUT_PATH, "ERGMS")

# Recreate the subdirectories
os.makedirs(OUTPUT_PATH_ERGMS)

[162]: # Define the directory containing the .gml files
directory = 'assignment_03_data'

# Use glob to get a list of all .gml files in the directory
gml_files = glob.glob(os.path.join(directory, '*.gml'))

# Init an empty dictionary for all graphs - the key is the actual name, and the
↳ value is the graph data
graphs = {}

# Use tqdm to show a progress bar in Jupyter Notebook
with tqdm(total=len(gml_files), desc="Loading GML Files") as pbar:
    # Loop through the list of .gml files and load them
    for file_path in gml_files:
        with open(file_path, 'r') as file:
            content = file.read()

            # Use regex to extract the desired part
            match = re.search(r'/(~/+)\.gml$', file_path)
            if match:
                graph_name = match.group(1)

                # Add to the graph dictionary
                graph = nx.read_gml(file_path)
                graphs[graph_name] = graph

            # Print the graph name when each part is finished
            print(f"Finished loading {graph_name}")

```

```
pbar.update(1) # Update the progress bar
```

Loading GML Files: 50%| | 1/2 [00:00<00:00, 8.08it/s]

Finished loading graph_eu_airlines

Loading GML Files: 100%| | 2/2 [00:00<00:00, 3.30it/s]

Finished loading graph_power

```
[163]: # Define the directory containing the .gml files
directory = 'assignment_03_data/World_Trade_Web'

# Use glob to get a list of all .gml files in the directory
gml_files = glob.glob(os.path.join(directory, '*.graphml'))

# Init an empty dictionary for all graphs - the key is the actual name, and the
# value is the graph data
graphs_wtn = {}

# Use tqdm to show a progress bar in Jupyter Notebook
with tqdm(total=len(gml_files), desc="Loading GML WTW Files") as pbar:
    # Loop through the list of .gml files and load them
    for file_path in gml_files:
        with open(file_path, 'r') as file:
            content = file.read()

            # Use regex to extract the desired part
            match = re.search(r'/(~/+)\.txt\.graphml$', file_path)
            if match:
                graph_name = match.group(1)

                # Add to the graph dictionary
                graph = nx.read_graphml(file_path)
                graphs_wtn[graph_name] = graph

                # Print the graph name when each part is finished
                print(f"Finished loading {graph_name}")

pbar.update(1) # Update the progress bar
```

Loading GML WTW Files: 9%| | 1/11 [00:00<00:08, 1.23it/s]

Finished loading WDN_1997

Finished loading WDN_1992

Loading GML WTW Files: 27%| | 3/11 [00:01<00:02, 3.30it/s]

Finished loading WDN_2000

```

Loading GML WTW Files: 45%|          | 5/11 [00:01<00:01, 3.26it/s]
Finished loading WDN_1998
Finished loading WDN_1994
Loading GML WTW Files: 64%|          | 7/11 [00:02<00:01, 3.33it/s]
Finished loading WDN_1993
Finished loading WDN_2002
Loading GML WTW Files: 82%|          | 9/11 [00:03<00:00, 3.40it/s]
Finished loading WDN_1996
Finished loading WDN_1995
Loading GML WTW Files: 91%|          | 10/11 [00:03<00:00, 2.71it/s]
Finished loading WDN_1999
Loading GML WTW Files: 100%|         | 11/11 [00:04<00:00, 2.62it/s]
Finished loading WDN_2001

```

2 I ERGMs

```
[164]: graphs
```

```
[164]: {'graph_eu_airlines': <networkx.classes.graph.Graph at 0x138de2dd0>,
      'graph_power': <networkx.classes.graph.Graph at 0x1434dbfd0>}
```

```
[165]: graphs_wtn
```

```
[165]: {'WDN_1997': <networkx.classes.digraph.DiGraph at 0x1445ad210>,
      'WDN_1992': <networkx.classes.digraph.DiGraph at 0x143cfaa10>,
      'WDN_2000': <networkx.classes.digraph.DiGraph at 0x142fadb90>,
      'WDN_1998': <networkx.classes.digraph.DiGraph at 0x14309b7d0>,
      'WDN_1994': <networkx.classes.digraph.DiGraph at 0x14327a410>,
      'WDN_1993': <networkx.classes.digraph.DiGraph at 0x143efa150>,
      'WDN_2002': <networkx.classes.digraph.DiGraph at 0x1451992d0>,
      'WDN_1996': <networkx.classes.digraph.DiGraph at 0x144142e50>,
      'WDN_1995': <networkx.classes.digraph.DiGraph at 0x14518a1d0>,
      'WDN_1999': <networkx.classes.digraph.DiGraph at 0x14552e490>,
      'WDN_2001': <networkx.classes.digraph.DiGraph at 0x145c8c090>}
```

2.1 1. Compute the strength assortativity coefficient on the data.

Hint: Pay attention: the underlying data (WTW) are weighted and directed, meaning you should account for four different kind of assortativity: in-in, in-out, out-in, out-out.

```
[166]: def calculate_assortativity(graph):
    in_in = nx.assortativity.degree_pearson_correlation_coefficient(G=graph,
↪x='in', y='in', weight='weight')
    in_out = nx.assortativity.degree_pearson_correlation_coefficient(G=graph,
↪x='in', y='out', weight='weight')
    out_in = nx.assortativity.degree_pearson_correlation_coefficient(G=graph,
↪x='out', y='in', weight='weight')
    out_out = nx.assortativity.degree_pearson_correlation_coefficient(G=graph,
↪x='out', y='out', weight='weight')

    # Create a dictionary to store assortativity values
    assortativity_dict = {
        "In-In Assortativity": in_in,
        "In-Out Assortativity": in_out,
        "Out-In Assortativity": out_in,
        "Out-Out Assortativity": out_out
    }

    # Return dictionary with assortativity values
    return assortativity_dict
```

```
[167]: assortativities = {}
for graph_name, graph in graphs_wtn.items():
    assortativities[graph_name] = calculate_assortativity(graph=graph)

print(assortativities)
```

```
{'WDN_1997': {'In-In Assortativity': -0.07368635207387701, 'In-Out
Assortativity': -0.07132955497657792, 'Out-In Assortativity':
-0.07171683897072376, 'Out-Out Assortativity': -0.06970079872009399},
'WDN_1992': {'In-In Assortativity': -0.04868776409103533, 'In-Out
Assortativity': -0.046647404340923876, 'Out-In Assortativity':
-0.059869793086991335, 'Out-Out Assortativity': -0.05760318920541264},
'WDN_2000': {'In-In Assortativity': -0.0631970717497539, 'In-Out Assortativity':
-0.06467170213846346, 'Out-In Assortativity': -0.06539073425903164, 'Out-Out
Assortativity': -0.06730072607061888}, 'WDN_1998': {'In-In Assortativity':
-0.0655705781839572, 'In-Out Assortativity': -0.06540616504184192, 'Out-In
Assortativity': -0.06572057397949527, 'Out-Out Assortativity':
-0.06583617855060356}, 'WDN_1994': {'In-In Assortativity': -0.07868988970611883,
'In-Out Assortativity': -0.07622523439269807, 'Out-In Assortativity':
-0.08022501617948047, 'Out-Out Assortativity': -0.07800143058821393},
'WDN_1993': {'In-In Assortativity': -0.06537929984057925, 'In-Out
Assortativity': -0.062189867187614475, 'Out-In Assortativity':
-0.07202068872957715, 'Out-Out Assortativity': -0.06910618411247677},
'WDN_2002': {'In-In Assortativity': -0.06019317176336694, 'In-Out
Assortativity': -0.060795564663496796, 'Out-In Assortativity':
-0.06303814946654555, 'Out-Out Assortativity': -0.06417872159686847},
```

```
'WDN_1996': {'In-In Assortativity': -0.07686964594098099, 'In-Out Assortativity': -0.07546909805023332, 'Out-In Assortativity': -0.07535502512416227, 'Out-Out Assortativity': -0.07420828746119328},
'WDN_1995': {'In-In Assortativity': -0.07921393553292572, 'In-Out Assortativity': -0.07702100817504068, 'Out-In Assortativity': -0.07830645801950578, 'Out-Out Assortativity': -0.07638722326404519},
'WDN_1999': {'In-In Assortativity': -0.06360116568177393, 'In-Out Assortativity': -0.06507198196482604, 'Out-In Assortativity': -0.06546711179930208, 'Out-Out Assortativity': -0.06730720421208614},
'WDN_2001': {'In-In Assortativity': -0.060016931181439886, 'In-Out Assortativity': -0.061178239732267325, 'Out-In Assortativity': -0.06192625961852602, 'Out-Out Assortativity': -0.06348734473716487}}
```

```
[168]: # Extracting graph names and assortativity values
graph_names = list(assortativities.keys())
keys = list(assortativities[graph_names[0]].keys()) # Extracting keys for
↳ assortativity values
values = {graph_name: [assortativities[graph_name][key] for key in keys] for
↳ graph_name in graph_names}

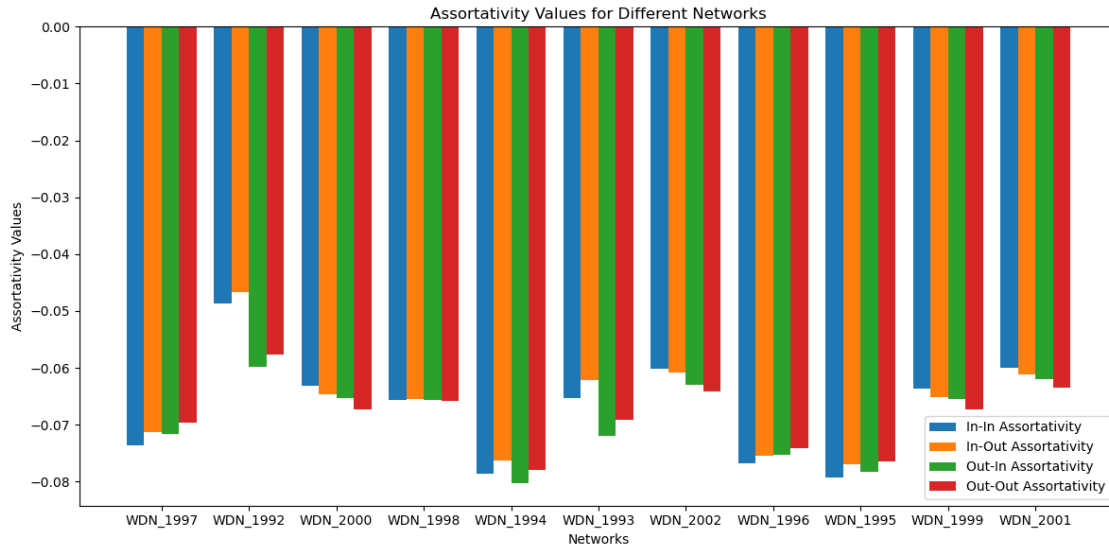
# Creating side-by-side bar plots for each network
bar_width = 0.2 # Width of each bar
index = np.arange(len(graph_names)) # Index for the networks

plt.figure(figsize=(12, 6))

for i, key in enumerate(keys):
    plt.bar(index + (i * bar_width), [values[graph_name][i] for graph_name in
↳ graph_names], bar_width, label=key)

plt.xlabel('Networks')
plt.ylabel('Assortativity Values')
plt.title('Assortativity Values for Different Networks')
plt.xticks(index + (bar_width * (len(keys) - 1) / 2), graph_names)
plt.legend()
plt.tight_layout()

plt.show()
```



2.2 2. Fit the Undirected Enhanced CM and Directed Enhanced CM using the CReMa method.

Hint: You can use the `nentropy` package on `networkx` to deploy these methods. You can find some examples at <https://github.com/nicoloval/NEMtropy/blob/master/examples/Directed%20Graphs>. `ipynb`. For additional questions, do not hesitate to ask at the QA.

```
[169]: # Create undirected versions of the graph
```

```
graphs_wtn_undirected = {}
for graph_name, graph in graphs_wtn.items():
    graph_undirected = graph.to_undirected()

    graphs_wtn_undirected[graph_name] = graph_undirected
```

```
[170]: FIG_SIZE = (16, 12)
        NODE_SIZE = 100
        MAX_STEPS = 10000
```

```
[171]: graphs_wtn
```

```
[171]: {'WDN_1997': <networkx.classes.digraph.DiGraph at 0x1445ad210>,
        'WDN_1992': <networkx.classes.digraph.DiGraph at 0x143cfaa10>,
        'WDN_2000': <networkx.classes.digraph.DiGraph at 0x142fadb90>,
        'WDN_1998': <networkx.classes.digraph.DiGraph at 0x14309b7d0>,
        'WDN_1994': <networkx.classes.digraph.DiGraph at 0x14327a410>,
        'WDN_1993': <networkx.classes.digraph.DiGraph at 0x143efa150>,
        'WDN_2002': <networkx.classes.digraph.DiGraph at 0x1451992d0>,
```

```
'WDN_1996': <networkx.classes.digraph.DiGraph at 0x144142e50>,
'WDN_1995': <networkx.classes.digraph.DiGraph at 0x14518a1d0>,
'WDN_1999': <networkx.classes.digraph.DiGraph at 0x14552e490>,
'WDN_2001': <networkx.classes.digraph.DiGraph at 0x145c8c090>}
```

```
[172]: graphs_wtn_undirected
```

```
[172]: {'WDN_1997': <networkx.classes.graph.Graph at 0x145cbe390>,
'WDN_1992': <networkx.classes.graph.Graph at 0x145ce53d0>,
'WDN_2000': <networkx.classes.graph.Graph at 0x143eea850>,
'WDN_1998': <networkx.classes.graph.Graph at 0x146490bd0>,
'WDN_1994': <networkx.classes.graph.Graph at 0x14517ebd0>,
'WDN_1993': <networkx.classes.graph.Graph at 0x1448edc10>,
'WDN_2002': <networkx.classes.graph.Graph at 0x1448f4b10>,
'WDN_1996': <networkx.classes.graph.Graph at 0x143d006d0>,
'WDN_1995': <networkx.classes.graph.Graph at 0x142f61510>,
'WDN_1999': <networkx.classes.graph.Graph at 0x142b5eed0>,
'WDN_2001': <networkx.classes.graph.Graph at 0x1429ac2d0>}
```

```
[173]: nemtropy_directed_results = {}
nemtropy_undirected_results = {}
```

```
[174]: def to_entropy_graph(graph):
    # Extract necessary information from the NetworkX graph
    adj_weigh = nx.to_numpy_array(graph) # Adjacency matrix
    edges = list(graph.edges()) # Edge list
    degrees = dict(graph.degree()) # Degree sequence (as a dictionary)
    strengths = dict(graph.degree(weight='weight')) # Strength sequence
    ↪(assuming weights are present)

    return adj_weigh, edges, degrees, strengths
```

```
[175]: def to_networkx_graph(adj_weigh):
    graph = nx.from_numpy_array(adj_weigh, create_using=nx.DiGraph)

    # Set edge attributes (weights) from the adjacency matrix
    edges = graph.edges(data=True)
    for (u, v, d) in edges:
        d['weight'] = adj_weigh[u][v]
    return graph
```

```
[176]: def plot_networkx(graph, graph_name, is_directed):

    # Plot the graph
    plt.figure(figsize=FIG_SIZE)

    # Set the layout for consistent plotting
```

```

pos = nx.kamada_kawai_layout(graph)

# Draw nodes and labels
nx.draw(graph, pos, with_labels=True, node_size=NODE_SIZE,
↪node_color='skyblue', font_weight='bold')

# Draw edge labels with weights
edge_labels = nx.get_edge_attributes(graph, 'weight')
nx.draw_networkx_edge_labels(graph, pos, edge_labels=edge_labels)

plt.title(f'{ "Undirected" if not is_directed else "Directed" } Weighted_
↪Graph of Network {graph_name} (networkx plot)')
file_name=f'{ "undirected_" if not is_directed else_
↪"directed_" }weighted_graph_{graph_name}_networkx.png'
complete_plot_filename=os.path.join(OUTPUT_PATH_ERGMS, file_name)
plt.savefig(complete_plot_filename)
plt.show()

```

```

[177]: def plot_entropy_graph(adj_weigh, graph_name, is_directed):
# Create a NetworkX graph from the weighted adjacency matrix
graph = nx.from_numpy_array(adj_weigh, create_using=nx.DiGraph)

# Set edge attributes (weights) from the adjacency matrix
edges = graph.edges(data=True)
for (u, v, d) in edges:
    d['weight'] = adj_weigh[u][v]

# Plot the graph
plt.figure(figsize=FIG_SIZE)

pos = nx.spring_layout(graph, seed=SEED) # Set the layout for consistent_
↪plotting

# Draw nodes and labels
nx.draw(graph, pos, with_labels=True, node_size=NODE_SIZE,
↪node_color='skyblue', font_weight='bold')

# Draw edge labels with weights
edge_labels = nx.get_edge_attributes(graph, 'weight')
nx.draw_networkx_edge_labels(graph, pos, edge_labels=edge_labels)

plt.title(f'{ "Undirected" if not is_directed else "Directed" } Weighted_
↪Graph of Network {graph_name} (networkx plot)')
file_name=f'{ "undirected_" if not is_directed else_
↪"directed_" }weighted_graph_{graph_name}_networkx.png'
complete_plot_filename=os.path.join(OUTPUT_PATH_ERGMS, file_name)

```

```
plt.savefig(complete_plot_filename)
plt.show()
```

```
[178]: for graph_name, graph in graphs_wtn.items():

    # Plot the networkx graph
    # plot_networkx(graph=graph, graph_name=graph_name, is_directed=True)

    # Extract necessary information from the NetworkX graph
    adj_weigh, edges, degrees, strengths = to_entropy_graph(graph)

    # Instantiate DirectedGraph with the extracted information
    nemtropy_graph = DirectedGraph(
        adjacency=adj_weigh,
        edgelist=edges,
        degree_sequence=np.array(list(degrees.values())),
        strength_sequence=np.array(list(strengths.values()))
    )

    # Plot the entropy graph
    # plot_entropy_graph(adj_weigh=adj_weigh, graph_name=graph_name,
    ↪is_directed=True)

    # We can the see the graph sequence of the graph by tapping
    print("Out strength sequence ", nemtropy_graph.out_strength)
    print("In strength sequence ", nemtropy_graph.in_strength)

    # Create the fit
    fit = nemtropy_graph.solve_tool(model="crema",
        method="newton",
        initial_guess="random",
        adjacency="dcm_exp",
        method_adjacency="newton",
        max_steps=MAX_STEPS)

    nemtropy_directed_results[graph_name] = {'nemtropy_graph': nemtropy_graph ↪
    ↪, 'fit': fit, 'error_degree': nemtropy_graph.error_degree, 'error_strength': ↪
    ↪nemtropy_graph.error_strength}
    print(nemtropy_graph.error_degree, nemtropy_graph.error_strength)
```

```
Out strength sequence [2.50279029e+08 1.51110308e+10 4.61846567e+09
6.81711200e+07
2.67190749e+10 2.21502936e+08 5.41374088e+10 4.91168703e+10
7.49909115e+08 7.29857541e+08 1.87668938e+09 4.48724200e+09
2.25759191e+08 5.02600422e+09 1.29821821e+11 2.60318865e+08
5.41707901e+08 2.26150610e+07 1.02480718e+09 5.25837704e+10
2.82353797e+09 4.23966182e+09 2.35580441e+08 9.52593690e+07
5.48710564e+08 2.44952892e+09 1.13775298e+11 3.16235040e+07
```

1.85608125e+08 2.50753772e+08 1.64791722e+10 2.04789199e+11
1.21519771e+10 1.37451310e+07 2.07905204e+09 4.64450332e+09
4.46854741e+09 3.05960474e+09 7.69311960e+08 1.28407149e+09
4.06837602e+10 1.93593910e+07 8.93722920e+07 5.04968621e+09
5.98755785e+09 6.02622009e+09 2.46597158e+09 3.91298119e+08
5.33239834e+08 3.60664557e+10 5.37243648e+08 2.49628523e+11
3.44346969e+09 1.78317524e+08 3.95918789e+11 1.43725953e+09
9.81569645e+09 3.22718120e+07 3.94739741e+09 8.39709792e+08
6.55107780e+07 5.97552384e+08 2.41601567e+08 3.36166761e+09
5.11124305e+10 1.86527725e+10 1.98075262e+09 3.93870803e+10
4.81566619e+10 1.71779386e+10 4.76780112e+10 2.12533630e+10
2.05959745e+11 1.89375834e+09 2.65241639e+11 7.91669369e+08
5.63500059e+09 1.76094056e+09 3.86402060e+07 1.05361959e+11
1.19883308e+10 3.85510096e+08 2.47610010e+08 3.00390106e+09
5.69702382e+08 5.86877248e+09 2.90121639e+09 7.01024071e+08
5.66601608e+08 6.56736261e+10 8.64464800e+07 5.30174574e+08
6.73943772e+08 1.65367918e+09 5.53708024e+10 1.26426345e+09
4.72318069e+08 7.37043076e+09 3.05583672e+08 1.12033514e+09
4.50691711e+08 1.60240152e+11 1.38063194e+10 8.14404405e+08
2.70075816e+08 1.26978910e+10 4.22322806e+10 6.67291438e+09
9.77470426e+09 2.87636430e+09 2.20068310e+09 1.54066303e+09
6.65915080e+09 2.40319259e+10 2.19073634e+10 2.32562064e+10
5.54558377e+09 8.09868187e+09 6.71526368e+10 8.97138000e+07
5.21120300e+07 1.05806497e+08 1.66888360e+08 6.03213340e+07
1.76544620e+07 3.26313975e+10 6.59419990e+08 1.03333214e+08
2.03148481e+08 6.87598290e+10 7.51085120e+09 2.68097587e+10
2.33235136e+08 9.18797086e+10 4.22557602e+09 5.65629210e+08
5.51530478e+08 7.83342022e+10 8.71373641e+10 4.07995601e+09
4.23721726e+08 7.56132046e+08 5.70826279e+10 4.17227193e+08
1.67149090e+07 2.46907814e+09 5.43612805e+09 2.22917101e+10
1.52490704e+09 6.34142339e+08 1.13109486e+10 1.79301992e+10
2.24632146e+11 2.98043026e+09 4.95992466e+11 4.41596769e+09
7.45763990e+07 1.48813943e+10 8.75361504e+09 2.37198168e+09
8.75424775e+08 2.13156288e+09]
In strength sequence [6.19525551e+08 8.67004516e+09 0.00000000e+00
0.00000000e+00
2.92027283e+10 7.07878341e+08 5.85864132e+10 5.68641270e+10
7.53695469e+08 1.57640498e+09 0.00000000e+00 7.32995147e+09
9.80107808e+08 0.00000000e+00 1.48684922e+11 2.57792318e+08
0.00000000e+00 0.00000000e+00 1.90074366e+09 6.46255190e+10
2.09315054e+09 4.67431859e+09 7.21910618e+08 1.19666379e+08
0.00000000e+00 1.46277071e+09 1.13955493e+11 2.54931462e+08
9.28995320e+07 0.00000000e+00 1.77742586e+10 1.16950804e+11
1.51069490e+10 5.54385970e+07 0.00000000e+00 4.47681420e+09
2.73519740e+09 8.58618093e+09 3.45340240e+09 0.00000000e+00
4.25211998e+10 0.00000000e+00 1.31157000e+08 0.00000000e+00
4.37531957e+09 1.23056200e+10 2.92027317e+09 0.00000000e+00
0.00000000e+00 2.94429123e+10 1.08392351e+09 2.37690282e+11

```

9.82018041e+08 2.60553934e+08 3.59446611e+11 3.25989969e+09
2.65893874e+10 1.65614941e+08 3.69660625e+09 6.19658985e+08
0.00000000e+00 0.00000000e+00 0.00000000e+00 2.11926354e+09
1.49046660e+11 1.99733690e+10 1.98405064e+09 3.73515026e+10
4.07149924e+10 1.37815748e+10 3.75986585e+10 2.67544740e+10
1.93224364e+11 2.97225766e+09 2.67013883e+11 3.24478541e+09
4.21580948e+09 3.26216992e+09 3.87813070e+07 1.21883710e+11
0.00000000e+00 0.00000000e+00 0.00000000e+00 2.52531113e+09
7.16766344e+09 0.00000000e+00 5.39019025e+09 5.45947579e+08
0.00000000e+00 6.18559197e+10 3.58887696e+08 6.85890650e+08
0.00000000e+00 2.43704626e+09 7.67765617e+10 3.85501394e+08
4.63824044e+08 7.94668968e+09 6.78520734e+08 0.00000000e+00
0.00000000e+00 1.49625759e+11 1.41056888e+10 1.44749297e+09
3.53738296e+08 6.25782376e+09 3.46909909e+10 5.02300622e+09
0.00000000e+00 2.55131910e+09 0.00000000e+00 3.16618249e+09
8.47922578e+09 3.69653296e+10 3.98784767e+10 3.48450137e+10
0.00000000e+00 1.11062520e+10 4.97830882e+10 2.85806872e+08
1.42180169e+08 3.28393892e+08 1.80020818e+08 0.00000000e+00
0.00000000e+00 0.00000000e+00 1.62569526e+09 3.38116307e+08
0.00000000e+00 9.22847656e+10 8.87912132e+09 2.34516294e+10
0.00000000e+00 1.14354167e+11 0.00000000e+00 1.61793038e+09
0.00000000e+00 6.20065235e+10 7.52280550e+10 0.00000000e+00
0.00000000e+00 1.17755579e+09 5.87915266e+10 6.51116167e+08
0.00000000e+00 3.01381587e+09 8.39593539e+09 4.76032455e+10
1.16982376e+09 7.93822326e+08 1.23191050e+10 0.00000000e+00
2.47730624e+11 3.64733113e+09 5.05568370e+11 0.00000000e+00
0.00000000e+00 1.29444472e+10 0.00000000e+00 0.00000000e+00
0.00000000e+00 0.00000000e+00]

```

```

/usr/local/anaconda3/envs/network-science/lib/python3.11/site-
packages/numba/core/utils.py:643: NumbaExperimentalFeatureWarning: First-class
function type feature is experimental
    warnings.warn("First-class function type feature is experimental",

```

```

solution error = 73029.06451416016
3.84389409191499e-09 73029.06451416016
Out strength sequence [5.64164710e+07 6.24782338e+09 2.88956492e+09
2.86005440e+07
9.33957273e+09 6.88719500e+06 2.10538162e+10 3.03846311e+10
6.93597580e+07 9.34456280e+08 7.92370638e+08 1.41043480e+09
8.16695620e+07 2.65861489e+08 6.83905357e+10 1.20770037e+08
8.31365780e+07 3.51032200e+06 3.84543557e+08 2.30325495e+10
8.92662122e+08 1.42609327e+09 1.10649754e+08 5.88415860e+07
1.58608416e+08 8.63400508e+08 5.96202792e+10 4.50548200e+06
2.37063350e+07 1.09317145e+08 5.89295619e+09 5.72257013e+10
5.53990945e+09 1.57934980e+07 8.31126212e+08 2.11459480e+09
1.43185694e+09 8.07705137e+08 3.06135906e+08 5.00914086e+08
2.17158479e+10 6.01044600e+06 6.11580300e+07 2.59085332e+09

```

2.97988588e+09 2.97012301e+09 5.22737220e+08 2.97001310e+07
 2.50054620e+08 1.45721531e+10 7.75229410e+07 1.19649882e+11
 1.51565304e+09 1.15297980e+07 1.67041741e+11 6.00078443e+08
 4.60135770e+09 1.63824330e+07 1.55330203e+09 3.57970699e+08
 1.43768400e+07 2.07048073e+08 1.23541194e+08 1.13708998e+09
 4.68612924e+10 4.84400140e+09 8.00822695e+08 1.18908533e+10
 1.49987973e+10 8.53317335e+09 1.16898751e+10 7.34341002e+09
 1.00108540e+11 9.67159176e+08 1.65847791e+11 5.74434073e+08
 2.41352214e+08 5.10732173e+08 1.07553400e+06 3.64998137e+10
 2.32887692e+09 2.77544980e+07 7.16849550e+07 3.80331112e+08
 3.05323651e+08 4.90726985e+09 4.03344072e+08 1.54971586e+08
 3.11493294e+08 2.63563394e+10 6.19564520e+07 2.76492240e+08
 8.28581400e+07 4.37737103e+08 2.50682597e+10 1.63355835e+08
 6.33873180e+07 2.53323329e+09 2.46616686e+08 6.32457356e+08
 3.37366892e+08 6.59671425e+10 6.19987579e+09 2.04076010e+08
 1.49138730e+07 6.48774346e+09 1.91277754e+10 2.57279501e+09
 4.76638726e+09 1.96203460e+09 1.45952740e+09 7.29643757e+08
 2.39222906e+09 7.44721572e+09 8.96928473e+09 1.04837433e+10
 1.14581722e+09 2.00383304e+09 2.10958895e+10 5.52804590e+07
 2.58945540e+07 3.55035900e+07 2.96516000e+07 3.04738760e+07
 3.72741000e+06 2.38795540e+10 2.21574389e+08 3.10691010e+07
 1.51123620e+08 2.95394323e+10 2.43749312e+09 6.79523444e+09
 4.75635410e+07 3.00188438e+10 1.63148915e+09 2.70055791e+08
 2.12815511e+08 3.20847095e+10 3.79937292e+10 1.35439830e+09
 2.21474760e+07 2.92363151e+08 1.94467394e+10 1.42479550e+08
 1.03985160e+07 1.27852035e+09 1.59140995e+09 8.94109993e+09
 8.68542620e+07 1.39715351e+08 7.71751690e+08 7.47097260e+09
 1.10168483e+11 1.19928093e+09 2.25035521e+11 1.73139753e+08
 2.14698100e+07 7.59989931e+09 1.13865882e+09 6.40109808e+08
 6.92195062e+08 9.88960442e+08]

In strength sequence [0.00000000e+00 8.57305062e+09 0.00000000e+00
 0.00000000e+00

0.00000000e+00 0.00000000e+00 3.83777342e+10 0.00000000e+00
 0.00000000e+00 0.00000000e+00 0.00000000e+00 3.57201271e+09
 0.00000000e+00 0.00000000e+00 0.00000000e+00 2.69717732e+08
 0.00000000e+00 0.00000000e+00 1.11829837e+09 2.21146646e+10
 1.45862431e+09 0.00000000e+00 0.00000000e+00 0.00000000e+00
 0.00000000e+00 0.00000000e+00 7.94866732e+10 0.00000000e+00
 0.00000000e+00 0.00000000e+00 9.17894669e+09 7.52176806e+10
 6.48391441e+09 0.00000000e+00 0.00000000e+00 0.00000000e+00
 0.00000000e+00 4.41547342e+09 3.24309677e+09 0.00000000e+00
 3.20040563e+10 0.00000000e+00 0.00000000e+00 0.00000000e+00
 2.44661707e+09 0.00000000e+00 0.00000000e+00 0.00000000e+00
 0.00000000e+00 2.05613774e+10 0.00000000e+00 0.00000000e+00
 0.00000000e+00 0.00000000e+00 3.40859870e+11 0.00000000e+00
 2.31145924e+10 0.00000000e+00 0.00000000e+00 0.00000000e+00
 0.00000000e+00 3.30420869e+08 0.00000000e+00 0.00000000e+00
 0.00000000e+00 1.02544130e+10 1.64257315e+09 2.17060169e+10

```

2.65616143e+10 0.00000000e+00 2.22443179e+10 0.00000000e+00
0.00000000e+00 1.67839867e+09 0.00000000e+00 0.00000000e+00
0.00000000e+00 1.35651250e+09 0.00000000e+00 7.25022580e+10
0.00000000e+00 0.00000000e+00 0.00000000e+00 0.00000000e+00
0.00000000e+00 0.00000000e+00 0.00000000e+00 3.83962584e+08
0.00000000e+00 3.77556971e+10 0.00000000e+00 0.00000000e+00
0.00000000e+00 0.00000000e+00 4.48194960e+10 0.00000000e+00
0.00000000e+00 0.00000000e+00 0.00000000e+00 6.18307391e+08
0.00000000e+00 1.26608770e+11 8.92985070e+09 0.00000000e+00
0.00000000e+00 0.00000000e+00 0.00000000e+00 3.77031652e+09
0.00000000e+00 0.00000000e+00 0.00000000e+00 1.34831729e+09
3.84841315e+09 0.00000000e+00 0.00000000e+00 3.10095457e+10
0.00000000e+00 5.95473264e+09 0.00000000e+00 0.00000000e+00
0.00000000e+00 2.94509103e+08 0.00000000e+00 0.00000000e+00
0.00000000e+00 3.24826452e+10 0.00000000e+00 0.00000000e+00
0.00000000e+00 6.47758577e+10 0.00000000e+00 1.54850559e+10
0.00000000e+00 9.81252501e+10 3.40845719e+09 0.00000000e+00
0.00000000e+00 4.87532896e+10 6.48884339e+10 0.00000000e+00
0.00000000e+00 0.00000000e+00 3.88390636e+10 0.00000000e+00
0.00000000e+00 1.35386825e+09 6.69975834e+09 2.26142066e+10
0.00000000e+00 0.00000000e+00 0.00000000e+00 5.00152285e+09
0.00000000e+00 0.00000000e+00 3.42170961e+11 0.00000000e+00
0.00000000e+00 0.00000000e+00 0.00000000e+00 0.00000000e+00
0.00000000e+00 0.00000000e+00]

```

solution error = 345585.3070678711

3.1084255169844255e-09 345585.3070678711

Out strength sequence [2.84181095e+08 1.86387143e+10 7.88067957e+09

1.34623554e+08

```

2.59912721e+10 2.48191244e+08 5.74219128e+10 5.24104622e+10
1.35080611e+09 8.09297500e+08 2.77487166e+09 5.98807283e+09
2.25102966e+08 6.04196065e+09 2.63115817e+10 3.19966874e+08
4.42176983e+08 2.78886990e+07 1.01291793e+09 5.71283721e+10
3.73184030e+09 4.44066968e+09 2.93794228e+08 5.29793760e+07
1.46162805e+09 2.22220699e+09 1.18303962e+11 2.12131590e+07
2.46274707e+08 1.53548153e+08 1.88748568e+10 2.68140992e+11
1.42437666e+10 2.15416050e+07 1.98882172e+09 7.27372999e+09
4.16772493e+09 2.99253163e+09 9.75683972e+08 1.18546664e+09
4.04046196e+10 1.34618610e+07 9.56471510e+07 5.06787706e+09
6.29000997e+09 6.07232052e+09 3.13146385e+09 1.24168878e+09
5.72372531e+08 4.12788561e+10 3.10603554e+08 2.18109568e+11
3.56293340e+09 4.81912610e+07 3.69924079e+11 1.37140520e+09
9.47453191e+09 6.15667580e+07 4.84372021e+09 8.61563443e+08
1.33743164e+08 6.19247378e+08 3.38874756e+08 4.13884209e+09
5.77606959e+10 2.67353874e+10 2.16053412e+09 4.36745657e+10
5.70177053e+10 1.93078432e+10 5.94271361e+10 2.39040270e+10
1.91788775e+11 1.63371929e+09 2.69407448e+11 1.04996285e+09
8.48929378e+09 1.53220621e+09 2.03098730e+07 1.25408547e+11

```

1.32992414e+10 3.85532484e+08 2.29572098e+08 3.01118596e+09
 7.09948106e+08 9.45135895e+09 3.46786454e+09 9.43780122e+08
 4.13833832e+08 7.95864447e+10 1.48116854e+08 4.32589997e+08
 5.39312575e+08 1.62359308e+09 6.41725468e+10 8.28632196e+08
 4.48879326e+08 7.85590679e+09 3.80467635e+08 1.91077254e+09
 7.51686883e+08 1.38490351e+11 1.37014248e+10 1.02727399e+09
 4.00062335e+08 1.34076018e+10 4.19409753e+10 9.61531718e+09
 1.02847929e+10 2.44760352e+09 2.01702969e+09 1.19296121e+09
 6.21699370e+09 3.85186797e+10 2.71140511e+10 2.24599252e+10
 7.16401534e+09 1.01909071e+10 9.02285097e+10 5.67438640e+07
 7.77517910e+07 9.12007320e+07 1.08336266e+08 5.91438570e+07
 2.16699050e+07 4.10168353e+10 5.87069712e+08 1.86873958e+08
 1.55723355e+08 7.29099019e+10 7.69071463e+09 2.11306107e+10
 1.01157068e+08 9.78778341e+10 4.74107385e+09 1.63401377e+09
 5.52893657e+08 8.01925939e+10 8.82414929e+10 5.26355717e+09
 5.02949834e+08 6.73139407e+08 6.84764027e+10 3.39286057e+08
 1.98781680e+07 4.15653518e+09 5.65325034e+09 2.48807274e+10
 2.11103225e+09 3.58948448e+08 1.30379804e+10 1.79297724e+10
 2.20342347e+11 2.65000997e+09 4.94668134e+11 2.86037393e+09
 9.47683180e+07 1.27039102e+10 1.32323671e+10 3.21979615e+09
 6.03374115e+08 1.75415901e+09]
 In strength sequence [1.05616596e+09 8.92685242e+09 0.00000000e+00
 2.93206892e+08
 2.28921019e+10 7.34639607e+08 6.33110287e+10 5.53963889e+10
 1.15345536e+09 1.93596469e+09 4.70008726e+09 7.76502044e+09
 1.14817097e+09 8.27313527e+09 2.40108324e+10 4.02862776e+08
 5.69709234e+08 0.00000000e+00 1.78718326e+09 5.72532556e+10
 0.00000000e+00 6.32859120e+09 7.19844004e+08 1.48238669e+08
 1.26754374e+09 0.00000000e+00 1.45206304e+11 2.31562192e+08
 6.06208230e+07 0.00000000e+00 1.62510873e+10 1.66756008e+11
 1.13388740e+10 3.51297460e+07 0.00000000e+00 5.97642084e+09
 2.46434691e+09 7.28937006e+09 3.71436802e+09 0.00000000e+00
 4.15216339e+10 0.00000000e+00 1.44699604e+08 0.00000000e+00
 3.39049843e+09 0.00000000e+00 3.66401811e+09 0.00000000e+00
 0.00000000e+00 3.08983317e+10 1.23908896e+09 2.26383933e+11
 9.42326932e+08 1.94453036e+08 3.56765195e+11 2.78860078e+09
 2.85000766e+10 2.36331109e+08 4.72792106e+09 5.87769241e+08
 0.00000000e+00 5.64588396e+08 0.00000000e+00 2.42291803e+09
 1.49462642e+11 2.98298940e+10 2.30403240e+09 3.48591438e+10
 3.25062045e+10 1.29714781e+10 4.71948703e+10 3.01683900e+10
 1.93196639e+11 3.06236618e+09 2.65564909e+11 3.65490100e+09
 4.91312204e+09 2.75326690e+09 0.00000000e+00 1.16114312e+11
 7.01814207e+09 0.00000000e+00 0.00000000e+00 2.89290693e+09
 6.04639234e+09 0.00000000e+00 5.12562900e+09 9.00523473e+08
 0.00000000e+00 6.29701361e+10 4.03427376e+08 7.92962369e+08
 0.00000000e+00 2.23315251e+09 1.08308359e+11 7.60558836e+08
 6.28070410e+08 1.13962326e+10 4.72956487e+08 0.00000000e+00
 1.59548784e+09 1.37721032e+11 1.34973739e+10 1.67749189e+09

```

3.90895763e+08 5.54313625e+09 3.28463583e+10 5.03624620e+09
0.00000000e+00 2.88342204e+09 1.03207080e+09 2.15202545e+09
7.33424832e+09 3.18073299e+10 4.50576570e+10 3.89662588e+10
3.22826944e+09 1.24148050e+10 3.28935360e+10 0.00000000e+00
1.93180375e+08 3.52140300e+08 1.59757145e+08 0.00000000e+00
2.74333040e+07 2.94998550e+10 1.55339587e+09 0.00000000e+00
0.00000000e+00 9.24881239e+10 9.42729790e+09 1.23089703e+10
0.00000000e+00 1.31865358e+11 0.00000000e+00 1.63663697e+09
5.12522849e+08 6.42587000e+10 7.99736755e+10 0.00000000e+00
0.00000000e+00 1.41846783e+09 5.32411357e+10 3.27026692e+08
0.00000000e+00 3.26881325e+09 8.89204208e+09 5.23719448e+10
1.63895935e+09 9.19033365e+08 1.35527479e+10 0.00000000e+00
2.44777462e+11 3.43025465e+09 5.91465719e+11 0.00000000e+00
8.08090520e+07 1.40948029e+10 0.00000000e+00 0.00000000e+00
8.07419276e+08 0.00000000e+00]

```

solution error = 190784.88024902344

2.1081412171497504e-09 190784.88024902344

Out strength sequence [2.74442540e+08 1.18216521e+10 3.50157154e+09
2.82529570e+07

```

2.65011009e+10 2.25418464e+08 4.87807030e+10 5.23651976e+10
4.44349519e+08 5.83699628e+08 2.15261608e+09 4.81652704e+09
2.09025472e+08 1.98661807e+09 1.30795718e+11 2.94032397e+08
4.39231470e+08 1.46409710e+07 9.47779640e+08 5.19125081e+10
2.08512907e+09 4.16695064e+09 3.75106575e+08 7.24170660e+07
7.70776910e+08 2.09862162e+09 1.08267315e+11 3.12331440e+07
2.56580166e+08 2.43237414e+08 1.48954614e+10 2.03117324e+11
1.17948531e+10 1.18416690e+07 1.25997401e+09 5.71106818e+09
4.83162927e+09 3.08254114e+09 8.37799510e+08 1.11969220e+09
4.01801989e+10 2.59112290e+07 8.78312930e+07 5.11521290e+09
5.17169645e+09 5.37361532e+09 2.58621871e+09 4.47885578e+08
5.52413185e+08 3.80885574e+10 4.78005620e+08 2.49129205e+11
2.34969896e+09 1.55975543e+08 3.92417625e+11 1.66796720e+09
9.08208222e+09 4.34307740e+07 4.26936854e+09 8.88712954e+08
6.53256040e+07 5.68759867e+08 3.23685370e+08 3.72225152e+09
4.99782738e+10 2.26664458e+10 2.13450097e+09 3.83804017e+10
4.52160654e+10 1.17977475e+10 5.82599971e+10 2.25851586e+10
2.06463046e+11 1.88579782e+09 2.56899934e+11 1.00038513e+09
4.89735861e+09 1.78344803e+09 2.87573570e+07 1.05618086e+11
8.99644284e+09 5.01592714e+08 1.96396698e+08 2.73909278e+09
5.93515095e+08 7.25999250e+09 3.16279853e+09 7.70054969e+08
4.84951228e+08 6.24211783e+10 9.02771690e+07 5.41920676e+08
5.89101406e+08 1.72336859e+09 6.40099203e+10 1.02887813e+09
3.81376544e+08 7.72545294e+09 2.89151478e+08 1.04535381e+09
4.88610058e+08 1.66514953e+11 1.26693325e+10 8.47335711e+08
2.77249438e+08 7.31197957e+09 3.91470895e+10 4.64066767e+09
9.70948997e+09 2.39096232e+09 1.70858119e+09 1.38801290e+09
5.49200854e+09 2.64283670e+10 2.40364482e+10 2.35102542e+10

```

4.96556700e+09 8.67178608e+09 6.29405479e+10 8.21083090e+07
 5.00652020e+07 9.91420720e+07 1.40178057e+08 5.69937590e+07
 1.49636270e+07 3.15186966e+10 7.37446964e+08 1.12935336e+08
 1.19259962e+08 6.77882105e+10 8.05032268e+09 2.61279378e+10
 2.04168764e+08 9.94715898e+10 4.38344445e+09 6.77404327e+08
 5.74405247e+08 8.00034312e+10 8.94807996e+10 3.27614887e+09
 3.89648648e+08 7.35689770e+08 5.43070710e+10 3.29507973e+08
 1.58996170e+07 2.21007433e+09 5.81337420e+09 2.49327892e+10
 4.16392952e+08 4.84455283e+08 1.18698997e+10 1.69852077e+10
 2.25388001e+11 3.09483496e+09 4.99546618e+11 2.99059815e+09
 8.03388670e+07 1.29030396e+10 9.03352230e+09 1.65106861e+09
 7.92418827e+08 1.87304774e+09]

In strength sequence [8.34709535e+08 9.35052629e+09 0.00000000e+00
 0.00000000e+00

3.07784160e+10 0.00000000e+00 5.77242172e+10 5.99997059e+10
 1.03135414e+09 1.78813886e+09 0.00000000e+00 7.36184812e+09
 1.00914034e+09 8.40091935e+09 1.52425064e+11 2.69185120e+08
 8.37658775e+08 1.34679955e+08 2.33915529e+09 6.02675984e+10
 1.56024330e+09 4.81662172e+09 7.90414940e+08 1.61726819e+08
 0.00000000e+00 1.45417457e+09 1.18196104e+11 2.26777251e+08
 9.42146940e+07 0.00000000e+00 1.67661300e+10 1.13489968e+11
 1.43085034e+10 4.67914380e+07 0.00000000e+00 6.18607990e+09
 2.98541738e+09 7.80348881e+09 3.59100714e+09 0.00000000e+00
 4.38946982e+10 0.00000000e+00 0.00000000e+00 0.00000000e+00
 5.51729383e+09 1.47082826e+10 3.11914823e+09 0.00000000e+00
 0.00000000e+00 3.07428521e+10 1.43128441e+09 2.43305688e+11
 1.20556035e+09 2.65649995e+08 3.81790942e+11 2.76975436e+09
 2.97310916e+10 1.98614159e+08 4.54041845e+09 5.18767329e+08
 0.00000000e+00 0.00000000e+00 0.00000000e+00 2.51175183e+09
 1.37642690e+11 2.42837210e+10 2.42305806e+09 4.19048582e+10
 2.70660045e+10 1.39453672e+10 4.21281610e+10 2.56569920e+10
 1.95347377e+11 2.86999474e+09 2.19379932e+11 3.41561145e+09
 4.18716748e+09 3.25582969e+09 3.20799760e+07 8.06846679e+10
 0.00000000e+00 0.00000000e+00 0.00000000e+00 2.93465290e+09
 6.84696337e+09 0.00000000e+00 5.51315812e+09 5.02960508e+08
 0.00000000e+00 4.62667708e+10 3.63976354e+08 8.26641418e+08
 0.00000000e+00 2.41731347e+09 8.57202417e+10 1.00245857e+09
 4.82322611e+08 1.05708772e+10 0.00000000e+00 0.00000000e+00
 1.33307380e+09 1.38001515e+11 1.10023360e+10 1.52180213e+09
 4.24076863e+08 5.61079546e+09 3.65256216e+10 5.68044652e+09
 0.00000000e+00 2.88724800e+09 1.35355159e+09 2.67050836e+09
 8.20772601e+09 3.02541722e+10 4.36855900e+10 3.74478338e+10
 0.00000000e+00 1.15403630e+10 3.78866427e+10 2.92785820e+08
 0.00000000e+00 3.24696584e+08 1.89518241e+08 0.00000000e+00
 0.00000000e+00 2.93855827e+10 1.57578426e+09 3.82534727e+08
 0.00000000e+00 7.99169373e+10 9.56736508e+09 2.21270069e+10
 0.00000000e+00 1.23994317e+11 0.00000000e+00 2.01851554e+09
 0.00000000e+00 6.72233493e+10 7.93681883e+10 0.00000000e+00

0.00000000e+00 1.67507437e+09 4.02085645e+10 6.96095692e+08
0.00000000e+00 2.97617928e+09 8.86894837e+09 4.52274662e+10
9.45301381e+08 1.10147225e+09 9.87056515e+09 0.00000000e+00
2.50733981e+11 3.71345989e+09 5.37553694e+11 0.00000000e+00
0.00000000e+00 1.39943973e+10 0.00000000e+00 0.00000000e+00
1.09617473e+09 0.00000000e+00]

solution error = 225387.19116210938

1.9816468466160586e-09 225387.19116210938

Out strength sequence [1.64137767e+08 9.42802484e+09 3.05450477e+09

3.70233810e+07

1.55688972e+10 1.66268860e+07 3.93643738e+10 4.17133402e+10
1.06864025e+08 6.36556094e+08 1.42538619e+09 2.76820981e+09
1.35686722e+08 1.02837747e+09 1.07859215e+11 1.89481156e+08
2.60624972e+08 2.48365810e+07 8.32823561e+08 4.26358093e+10
2.44116486e+09 2.67976621e+09 1.30768351e+08 9.27555110e+07
2.41968651e+08 1.82684291e+09 9.30030207e+10 8.24082000e+06
5.03718500e+07 1.03650974e+08 1.16063061e+10 1.46054922e+11
8.29348554e+09 2.11947450e+07 9.55980377e+08 3.36061902e+09
2.81150027e+09 3.26049788e+09 6.27758245e+08 7.21758070e+08
3.21887649e+10 9.59584100e+06 1.05182517e+08 3.58953546e+09
4.46065773e+09 5.75077833e+09 1.34637904e+09 5.18725070e+07
5.18007900e+08 2.68178401e+10 2.52655549e+08 1.78993482e+11
2.43510308e+09 7.70551210e+07 3.01652078e+11 1.33158165e+09
7.51379105e+09 2.23160070e+07 2.60777227e+09 5.24342759e+08
8.18156860e+07 4.69147132e+08 8.32494840e+07 1.82183922e+09
4.55139125e+10 8.85886830e+09 1.68766032e+09 2.52613427e+10
3.32792769e+10 1.43089801e+10 2.89088391e+10 1.35798804e+10
1.59803737e+11 1.59887836e+09 2.24710236e+11 5.87889758e+08
9.57779217e+08 1.28198429e+09 1.00671700e+07 7.62982686e+10
8.56272494e+09 1.34953175e+08 1.88684717e+08 1.52193829e+09
4.37893217e+08 8.25979194e+09 1.34299172e+09 5.30823633e+08
3.91289302e+08 5.32426579e+10 6.42758650e+07 2.71239588e+08
4.18123803e+08 1.28645964e+09 3.52729122e+10 2.04181314e+08
2.06694442e+08 5.70649325e+09 2.30654308e+08 9.62083218e+08
4.01920244e+08 1.04688232e+11 1.12254513e+10 3.61044735e+08
1.08951770e+08 7.02186954e+09 3.19290610e+10 4.50242631e+09
7.91921627e+09 2.86016458e+09 2.69522008e+09 1.05582084e+09
4.19244540e+09 1.46054735e+10 1.42325715e+10 1.61731046e+10
2.94247584e+09 4.86279443e+09 4.52245457e+10 3.86188740e+07
3.54802500e+07 1.06543564e+08 9.41483580e+07 5.46970870e+07
7.06116100e+06 3.35120252e+10 4.16580962e+08 6.81319520e+07
1.45925937e+08 5.44594220e+10 5.73344586e+09 1.70778095e+10
1.62333172e+08 6.10083290e+10 2.84318704e+09 5.83380917e+08
3.74253581e+08 5.46159491e+10 6.85614975e+10 2.75520542e+09
3.84313544e+08 5.18756864e+08 4.02268624e+10 1.92315070e+08
2.10837950e+07 1.88221849e+09 4.07251695e+09 1.48695685e+10
6.48981494e+08 4.05659603e+08 5.02097175e+09 8.97556382e+09

```

1.65543315e+11 2.54809297e+09 3.90048610e+11 1.96004631e+09
2.94067630e+07 1.02854352e+10 4.20459211e+09 1.59378925e+09
9.13556023e+08 1.61326641e+09]
In strength sequence [0.00000000e+00 9.54664960e+09 0.00000000e+00
0.00000000e+00
2.13318898e+10 0.00000000e+00 4.74142028e+10 5.36519926e+10
0.00000000e+00 0.00000000e+00 0.00000000e+00 0.00000000e+00
0.00000000e+00 0.00000000e+00 0.00000000e+00 2.43080579e+08
0.00000000e+00 9.23662480e+07 1.18233164e+09 3.52935580e+10
1.84856107e+09 0.00000000e+00 0.00000000e+00 2.02576570e+08
0.00000000e+00 0.00000000e+00 9.63945468e+10 0.00000000e+00
1.16085217e+08 0.00000000e+00 1.08431121e+10 9.24329028e+10
1.17457810e+10 0.00000000e+00 3.18323141e+08 2.95592011e+09
0.00000000e+00 5.49997869e+09 2.83160188e+09 0.00000000e+00
3.23834786e+10 0.00000000e+00 9.31856620e+07 0.00000000e+00
3.53774676e+09 8.84925381e+09 2.24258842e+09 0.00000000e+00
0.00000000e+00 2.26305164e+10 0.00000000e+00 2.05662547e+11
6.57847303e+08 0.00000000e+00 3.23736954e+11 0.00000000e+00
2.13326848e+10 1.17452323e+08 2.56628700e+09 0.00000000e+00
0.00000000e+00 0.00000000e+00 0.00000000e+00 1.21207661e+09
1.24844967e+11 1.37442470e+10 1.45119632e+09 2.61350455e+10
3.10112717e+10 0.00000000e+00 2.43271348e+10 0.00000000e+00
1.57841140e+11 0.00000000e+00 2.30671162e+11 2.89519330e+09
0.00000000e+00 0.00000000e+00 0.00000000e+00 8.80620404e+10
0.00000000e+00 0.00000000e+00 0.00000000e+00 1.08032101e+09
0.00000000e+00 0.00000000e+00 2.49115470e+09 4.23870336e+08
5.48324764e+08 5.21712365e+10 0.00000000e+00 0.00000000e+00
0.00000000e+00 2.06722734e+09 6.09572270e+10 6.56639372e+08
0.00000000e+00 7.32404443e+09 4.85655000e+08 0.00000000e+00
1.13005734e+09 1.25585169e+11 1.15607834e+10 8.17297982e+08
0.00000000e+00 0.00000000e+00 2.70263970e+10 3.92376504e+09
0.00000000e+00 0.00000000e+00 0.00000000e+00 2.28239522e+09
5.54595273e+09 0.00000000e+00 2.03281320e+10 2.74362923e+10
0.00000000e+00 7.03012500e+09 0.00000000e+00 0.00000000e+00
1.22400035e+08 2.92490627e+08 1.27207730e+08 0.00000000e+00
0.00000000e+00 2.24932438e+10 0.00000000e+00 2.04545852e+08
0.00000000e+00 8.13635966e+10 6.97514597e+09 1.90767224e+10
0.00000000e+00 9.16147767e+10 4.50826743e+09 0.00000000e+00
5.40929715e+08 5.05305189e+10 6.74968484e+10 0.00000000e+00
0.00000000e+00 0.00000000e+00 5.12623454e+10 2.26384983e+08
0.00000000e+00 1.11641626e+09 6.86744240e+09 2.29068670e+10
0.00000000e+00 6.77629358e+08 0.00000000e+00 0.00000000e+00
1.83054474e+11 2.68067441e+09 4.06736886e+11 0.00000000e+00
7.77859000e+07 7.97127787e+09 0.00000000e+00 0.00000000e+00
0.00000000e+00 0.00000000e+00]

```

```

solution error = 206719.9835205078
5.512987399924896e-09 206719.9835205078

```

Out strength sequence [5.85536580e+07 6.84436185e+09 2.86572923e+09
3.20777430e+07

1.02141320e+10 1.88125420e+07 3.27708456e+10 3.04933478e+10
1.02099340e+08 8.47972840e+08 1.42302650e+09 1.86966265e+09
1.56228949e+08 4.17360922e+08 6.98424349e+10 1.86924855e+08
1.58661593e+08 2.47965700e+07 5.02282917e+08 3.11213512e+10
2.63483854e+09 1.67021470e+09 1.17934831e+08 6.23012580e+07
2.54562145e+08 1.00201816e+09 7.47761622e+10 3.48737800e+06
2.15205030e+07 9.64971350e+07 8.47665097e+09 1.14248928e+11
5.94817101e+09 1.42495330e+07 6.81479014e+08 2.54223251e+09
1.52663759e+09 1.34278554e+09 6.62632538e+08 5.63406513e+08
2.15557070e+10 5.91022000e+06 1.01159354e+08 2.96406375e+09
3.05649829e+09 3.41552710e+09 8.53050182e+08 3.87837890e+07
4.36069457e+08 1.86665632e+10 1.27712412e+08 1.36583325e+11
1.82537972e+09 7.04306030e+07 1.89992894e+11 8.53954282e+08
6.05777852e+09 2.40445720e+07 1.78954943e+09 4.18631156e+08
3.60672680e+07 3.99846299e+08 1.74939247e+08 1.42022546e+09
4.38184686e+10 4.86607661e+09 1.38393925e+09 2.01165502e+10
2.75968138e+10 1.23026187e+10 2.31993835e+10 1.03469587e+10
1.10009055e+11 1.38952644e+09 2.01298861e+11 5.51520819e+08
6.97292212e+08 8.19448497e+08 7.89538100e+06 6.20995516e+10
8.13648171e+09 8.03307850e+07 1.17878957e+08 8.87500973e+08
3.67654041e+08 3.96727469e+09 8.35611885e+08 2.26049171e+08
3.20271363e+08 4.06352578e+10 6.86157750e+07 2.86333665e+08
2.25113911e+08 9.17777261e+08 3.16771135e+10 1.51895798e+08
1.42847051e+08 2.74691733e+09 1.84943614e+08 8.65199929e+08
3.94604202e+08 6.71716083e+10 9.26747610e+09 2.21657959e+08
2.72458580e+07 6.17471072e+09 2.43996260e+10 4.38799228e+09
6.72544987e+09 2.28574660e+09 2.55053585e+09 8.83862449e+08
2.84613870e+09 1.18874075e+10 9.82212105e+09 1.37255154e+10
3.10009013e+09 2.53295835e+09 2.85444123e+10 6.36606090e+07
3.99041360e+07 1.26283698e+08 1.03843640e+08 5.55601160e+07
6.17937800e+06 2.56532618e+10 1.38386919e+08 4.11713480e+07
1.43135287e+08 4.64292156e+10 3.37876697e+09 1.20185421e+10
1.61375803e+08 3.12184208e+10 2.29551616e+09 3.68960802e+08
3.56320515e+08 3.97743400e+10 4.62648144e+10 1.85404225e+09
9.64372620e+07 4.39819153e+08 3.09296979e+10 1.04118250e+08
2.07711800e+07 1.38329156e+09 1.56687065e+09 1.02150090e+10
3.21468402e+08 1.24425088e+08 2.30761406e+09 8.65106103e+09
1.16990078e+11 1.92695174e+09 3.19900994e+11 6.95884308e+08
4.67529690e+07 8.72321073e+09 2.80470259e+09 1.03136876e+09
8.09615470e+08 1.04877304e+09]

In strength sequence [0.00000000e+00 8.73275160e+09 0.00000000e+00
0.00000000e+00

1.66519300e+10 0.00000000e+00 4.01617517e+10 0.00000000e+00
0.00000000e+00 0.00000000e+00 0.00000000e+00 3.70897250e+09
0.00000000e+00 0.00000000e+00 0.00000000e+00 2.72454218e+08
0.00000000e+00 9.19237150e+07 1.15867932e+09 2.74289501e+10

```

1.80764441e+09 0.00000000e+00 0.00000000e+00 2.18420756e+08
0.00000000e+00 0.00000000e+00 8.83566381e+10 0.00000000e+00
1.18786801e+08 0.00000000e+00 1.02452510e+10 8.11843889e+10
9.68256722e+09 0.00000000e+00 3.05129929e+08 0.00000000e+00
0.00000000e+00 4.83687440e+09 2.49504210e+09 0.00000000e+00
2.81465659e+10 0.00000000e+00 8.99558970e+07 0.00000000e+00
2.51224627e+09 0.00000000e+00 0.00000000e+00 0.00000000e+00
0.00000000e+00 1.76113563e+10 0.00000000e+00 0.00000000e+00
7.04899440e+08 0.00000000e+00 2.95860702e+11 0.00000000e+00
2.16674310e+10 1.23139294e+08 2.62501516e+09 0.00000000e+00
0.00000000e+00 0.00000000e+00 0.00000000e+00 0.00000000e+00
1.09871944e+11 1.08035870e+10 1.32524590e+09 2.07906416e+10
2.74643112e+10 0.00000000e+00 2.07170550e+10 0.00000000e+00
0.00000000e+00 2.04338147e+09 1.97554828e+11 0.00000000e+00
0.00000000e+00 0.00000000e+00 0.00000000e+00 7.45416879e+10
0.00000000e+00 0.00000000e+00 0.00000000e+00 0.00000000e+00
0.00000000e+00 0.00000000e+00 0.00000000e+00 4.62651352e+08
0.00000000e+00 3.89920367e+10 0.00000000e+00 0.00000000e+00
0.00000000e+00 2.03876541e+09 4.82181750e+10 0.00000000e+00
0.00000000e+00 6.75745014e+09 0.00000000e+00 0.00000000e+00
0.00000000e+00 1.11487964e+11 9.34923170e+09 7.29587108e+08
0.00000000e+00 0.00000000e+00 2.37377502e+10 4.11505510e+09
0.00000000e+00 0.00000000e+00 0.00000000e+00 1.57872913e+09
0.00000000e+00 0.00000000e+00 0.00000000e+00 2.44931309e+10
0.00000000e+00 6.49280513e+09 0.00000000e+00 0.00000000e+00
1.13367414e+08 2.90413073e+08 1.30368874e+08 0.00000000e+00
0.00000000e+00 2.74928123e+10 0.00000000e+00 0.00000000e+00
0.00000000e+00 7.30579845e+10 0.00000000e+00 1.57558256e+10
0.00000000e+00 7.83099231e+10 3.76485798e+09 0.00000000e+00
0.00000000e+00 4.17022516e+10 6.02640848e+10 0.00000000e+00
0.00000000e+00 0.00000000e+00 4.38319134e+10 0.00000000e+00
0.00000000e+00 1.37166338e+09 6.48670695e+09 2.92165135e+10
0.00000000e+00 0.00000000e+00 0.00000000e+00 5.31991606e+09
1.69409390e+11 0.00000000e+00 3.61438587e+11 0.00000000e+00
6.82305690e+07 0.00000000e+00 0.00000000e+00 0.00000000e+00
0.00000000e+00 0.00000000e+00]

```

solution error = 109617.70764160156

6.645926475812303e-09 109617.70764160156

Out strength sequence [3.59492905e+08 1.74056443e+10 7.18585544e+09

2.38456188e+08

```

2.64553843e+10 3.41694756e+08 5.70170018e+10 5.91067215e+10
1.30660963e+09 1.23409454e+09 2.53162882e+09 6.10315125e+09
3.15071889e+08 1.98396983e+09 2.26229572e+10 1.82604622e+08
2.97693475e+08 3.63672820e+07 1.25710396e+09 6.04985681e+10
3.26396161e+09 5.28231846e+09 2.22429173e+08 3.52494970e+07
1.88914101e+09 1.96791092e+09 1.06319907e+11 1.93612020e+07
1.96837361e+08 1.10060643e+08 1.79630099e+10 2.98846374e+11

```

1.28083287e+10 5.99652350e+07 1.69251200e+09 7.11623675e+09
 4.12612225e+09 3.08099894e+09 1.01213768e+09 1.41124971e+09
 4.50497087e+10 7.78956230e+07 5.90188910e+07 4.91245367e+09
 6.01025819e+09 6.77567472e+09 3.13968843e+09 1.86596924e+09
 4.87734624e+08 4.03868842e+10 3.74126323e+08 2.21024266e+11
 2.73100744e+09 3.39937460e+07 4.00518948e+11 1.51288795e+09
 1.02403871e+10 4.80159620e+07 4.91952182e+09 8.92529382e+08
 6.81281870e+07 5.30606811e+08 2.88013439e+08 4.27395142e+09
 5.02392642e+10 3.04145397e+10 2.32562117e+09 4.86268975e+10
 5.38830535e+10 1.39448089e+10 6.63074786e+10 2.18572614e+10
 2.06007073e+11 1.43684027e+09 2.40547824e+11 1.57558849e+09
 8.85297853e+09 1.94828908e+09 2.51444520e+07 1.15748025e+11
 1.40702125e+10 2.48792356e+08 1.55428814e+08 2.93585045e+09
 8.92497288e+08 6.29832212e+09 4.48862071e+09 8.46144275e+08
 4.59694512e+08 7.23678163e+10 2.10625607e+08 2.21557388e+08
 5.55947527e+08 1.67454817e+09 6.13371048e+10 8.03438559e+08
 5.05127701e+08 8.45308794e+09 9.30051562e+08 1.81565080e+09
 5.70295443e+08 1.48900658e+11 1.42170235e+10 1.11791081e+09
 1.00853840e+08 1.19470658e+10 4.23127395e+10 7.66488639e+09
 1.06046184e+10 2.86213105e+09 1.63650675e+09 1.28648615e+09
 6.59764520e+09 3.57729824e+10 3.36715039e+10 2.36153288e+10
 5.95678683e+09 1.38143808e+10 8.92347162e+10 1.07428801e+08
 6.91263340e+07 7.99237330e+07 1.68829380e+08 3.80390990e+07
 1.25396430e+07 3.65034895e+10 6.93441385e+08 3.60724479e+08
 1.47714607e+08 7.59091714e+10 8.07801126e+09 2.40047835e+10
 8.21120790e+07 1.05486293e+11 4.31747619e+09 2.08753796e+09
 4.69566067e+08 7.29058691e+10 9.16003034e+10 6.58428340e+09
 4.91013987e+08 6.54315864e+08 6.75385110e+10 2.39438455e+08
 2.87312490e+07 4.31719808e+09 6.44728651e+09 3.34427427e+10
 2.61614950e+09 3.54390294e+08 1.46958803e+10 2.27552056e+10
 2.09454876e+11 2.19503047e+09 4.67980436e+11 1.82902986e+09
 4.27006280e+07 1.06561680e+10 1.56485990e+10 2.09245020e+09
 5.83991556e+08 1.56293848e+09]

In strength sequence [1.47632923e+09 1.19791924e+10 0.00000000e+00
 0.00000000e+00

8.65053149e+09 8.32470594e+08 6.15638688e+10 5.91226598e+10
 1.64047173e+09 0.00000000e+00 3.41420346e+09 8.28906022e+09
 9.86827389e+08 8.75830703e+09 2.47368639e+10 3.31155460e+08
 7.33247417e+08 0.00000000e+00 1.72322136e+09 4.80242887e+10
 1.51981845e+09 7.48438415e+09 5.65515500e+08 1.27864381e+08
 1.49813368e+09 0.00000000e+00 1.46164720e+11 0.00000000e+00
 6.06336460e+07 0.00000000e+00 1.49928412e+10 1.90267416e+11
 1.26132656e+10 0.00000000e+00 0.00000000e+00 6.84395514e+09
 2.56858720e+09 9.10213511e+09 3.78120149e+09 0.00000000e+00
 4.60370914e+10 0.00000000e+00 1.12768287e+08 0.00000000e+00
 6.28779732e+09 1.09282108e+10 3.83836362e+09 0.00000000e+00
 7.75121015e+08 3.04976722e+10 1.55217247e+09 2.39560678e+11
 7.45015174e+08 1.59641026e+08 3.48915232e+11 0.00000000e+00

```

3.06994338e+10 1.96923398e+08 6.43188814e+09 6.52402622e+08
0.00000000e+00 4.95194806e+08 0.00000000e+00 2.89290000e+09
1.35783374e+11 3.48687640e+10 1.96550812e+09 3.95518044e+10
3.05175672e+10 1.96792430e+10 4.44987710e+10 2.68904380e+10
1.98215005e+11 3.44445489e+09 2.45526916e+11 4.09847981e+09
0.00000000e+00 2.96255480e+09 0.00000000e+00 1.25843240e+11
0.00000000e+00 5.75725527e+08 0.00000000e+00 3.66729039e+09
6.25179503e+09 0.00000000e+00 0.00000000e+00 5.02925823e+08
5.75815337e+08 6.52959564e+10 4.01913373e+08 0.00000000e+00
0.00000000e+00 2.24380165e+09 1.08842794e+11 1.00430172e+09
0.00000000e+00 1.19962456e+10 4.99368993e+08 0.00000000e+00
0.00000000e+00 1.35083387e+11 1.45626576e+10 1.75448546e+09
4.44641589e+08 8.18816600e+09 3.33974769e+10 5.98140020e+09
0.00000000e+00 2.51661726e+09 1.18322981e+09 1.65277360e+09
7.37853042e+09 2.95833349e+10 5.07272790e+10 3.90127647e+10
4.02126012e+09 1.73968970e+10 4.00757492e+10 2.30018043e+08
1.98587807e+08 3.13235354e+08 1.74446973e+08 1.31637606e+08
3.09323920e+07 3.13227686e+10 2.02162009e+09 2.45789807e+08
3.45060391e+08 9.28261309e+10 1.01653490e+10 1.17522954e+10
0.00000000e+00 1.45104772e+11 5.92758629e+09 2.29087635e+09
0.00000000e+00 6.31304286e+10 7.60426575e+10 4.13448350e+09
0.00000000e+00 1.56771790e+09 0.00000000e+00 4.00079264e+08
0.00000000e+00 0.00000000e+00 9.87531921e+09 4.85940592e+10
0.00000000e+00 1.04586908e+09 1.63024434e+10 0.00000000e+00
2.67753104e+11 1.94658849e+09 5.93215172e+11 0.00000000e+00
0.00000000e+00 1.12367997e+10 0.00000000e+00 0.00000000e+00
1.00927989e+09 1.72383985e+09]

```

solution error = 130727.1923828125

3.6602507691441133e-09 130727.1923828125

Out strength sequence [3.00936699e+08 1.38890365e+10 4.66968921e+09
5.56863780e+07

```

2.41094379e+10 1.04698144e+08 5.19219167e+10 4.92639905e+10
3.40264807e+08 6.79896369e+08 1.89531193e+09 3.90267985e+09
2.35882965e+08 4.67343134e+09 1.35332324e+11 2.54111196e+08
4.73371356e+08 4.11103480e+07 1.03028164e+09 4.79519341e+10
2.82850522e+09 3.79035224e+09 2.01377084e+08 9.88806840e+07
2.80035248e+08 2.27753221e+09 1.09916485e+11 1.71049890e+07
1.74481199e+08 2.17004609e+08 1.62711291e+10 1.82631824e+11
1.10171853e+10 2.17096970e+07 1.60850416e+09 4.11862746e+09
4.82794412e+09 3.37542826e+09 1.24640402e+09 1.60520502e+09
4.15259865e+10 1.83792520e+07 9.38534560e+07 4.23365370e+09
5.45694799e+09 6.26284799e+09 1.93294143e+09 1.92804178e+08
6.37476845e+08 3.48822402e+10 3.52986523e+08 2.41509442e+11
3.11493512e+09 1.74343330e+08 3.90477495e+11 1.50297206e+09
1.01940600e+10 2.26147280e+07 3.39985852e+09 8.54600431e+08
7.29239520e+07 5.57204034e+08 1.96372535e+08 2.71815536e+09
5.14899005e+10 1.51990959e+10 2.09433899e+09 3.65145632e+10

```

4.55393121e+10 1.88835664e+10 4.30998462e+10 1.91675848e+10
 2.10548978e+11 1.99852609e+09 2.55921529e+11 8.36897682e+08
 4.86733382e+09 1.81020745e+09 8.51265400e+06 9.64227749e+10
 1.19562108e+10 3.72657020e+08 2.76983269e+08 2.69690844e+09
 6.02315108e+08 6.43735789e+09 2.42942053e+09 6.90393835e+08
 4.50783695e+08 6.73328285e+10 6.88071390e+07 5.07320189e+08
 6.82065622e+08 1.63996384e+09 5.33934420e+10 1.23279284e+09
 4.13314166e+08 7.27643698e+09 2.54338676e+08 1.23158474e+09
 4.06561137e+08 1.62762460e+11 1.39082471e+10 6.66606960e+08
 9.76348650e+07 1.31605134e+10 4.22383927e+10 6.30143558e+09
 1.06398101e+10 3.17317578e+09 2.64989252e+09 1.55628041e+09
 5.52970155e+09 2.28402787e+10 2.05625029e+10 2.30796397e+10
 4.31045088e+09 7.35188068e+09 6.72642365e+10 6.25036740e+07
 3.86716270e+07 1.18259115e+08 9.88614470e+07 6.89734730e+07
 8.81108300e+06 2.96178167e+10 6.48389728e+08 6.38691150e+07
 2.09843392e+08 6.85777882e+10 7.55645143e+09 2.40430194e+10
 2.21570595e+08 8.91496676e+10 3.79119329e+09 5.96615702e+08
 5.72049974e+08 7.75971478e+10 8.82221489e+10 3.64761132e+09
 5.12647701e+08 7.57391044e+08 5.44889552e+10 3.24256990e+08
 1.85372420e+07 2.14670310e+09 5.44524272e+09 2.00673354e+10
 2.33905318e+09 6.39844552e+08 1.26078971e+10 1.46809250e+10
 2.33697577e+11 2.64050146e+09 4.66692774e+11 4.25687850e+09
 4.14723160e+07 1.33725015e+10 6.93351347e+09 2.05011840e+09
 9.63542050e+08 1.97116448e+09]

In strength sequence [9.17776890e+08 9.10608358e+09 0.00000000e+00
 0.00000000e+00

2.29739654e+10 0.00000000e+00 5.82032928e+10 5.95910019e+10
 9.29220321e+08 0.00000000e+00 0.00000000e+00 6.30708041e+09
 0.00000000e+00 0.00000000e+00 1.46930039e+11 2.30863947e+08
 0.00000000e+00 0.00000000e+00 1.63795728e+09 5.59400213e+10
 0.00000000e+00 4.58572216e+09 5.60031956e+08 1.37832286e+08
 0.00000000e+00 1.18692922e+09 1.01064592e+11 0.00000000e+00
 1.27007479e+08 0.00000000e+00 1.64557430e+10 1.13952785e+11
 1.34860594e+10 5.68874570e+07 0.00000000e+00 3.51096262e+09
 2.59391136e+09 7.41196109e+09 3.70928975e+09 0.00000000e+00
 3.89466411e+10 0.00000000e+00 1.27453750e+08 0.00000000e+00
 3.65116393e+09 1.22732435e+10 2.66043401e+09 0.00000000e+00
 0.00000000e+00 2.94361449e+10 0.00000000e+00 2.45706155e+11
 8.88117213e+08 2.24194900e+08 3.80951677e+11 2.41774715e+09
 2.80875204e+10 1.49486061e+08 3.00526356e+09 6.54655939e+08
 0.00000000e+00 0.00000000e+00 0.00000000e+00 1.74366692e+09
 1.41638361e+11 1.52223490e+10 2.01191653e+09 3.42391596e+10
 4.20237477e+10 0.00000000e+00 3.35119836e+10 2.74993640e+10
 1.93464327e+11 2.78823412e+09 2.70888003e+11 0.00000000e+00
 4.16416385e+09 0.00000000e+00 3.81446720e+07 1.21961224e+11
 0.00000000e+00 8.09584828e+08 0.00000000e+00 2.15322430e+09
 0.00000000e+00 0.00000000e+00 4.35882471e+09 5.77805964e+08
 0.00000000e+00 6.14497395e+10 3.07141074e+08 7.91666429e+08

```

0.00000000e+00 2.48953732e+09 5.87705704e+10 1.04961535e+09
4.26959188e+08 8.40096465e+09 7.20666000e+08 0.00000000e+00
0.00000000e+00 1.48515447e+11 1.43040334e+10 1.05087347e+09
3.40116846e+08 5.21986658e+09 3.38109597e+10 4.57165386e+09
0.00000000e+00 2.35197952e+09 0.00000000e+00 2.89476746e+09
7.87396627e+09 3.33436362e+10 3.50885470e+10 3.46361604e+10
0.00000000e+00 1.12315300e+10 4.33797150e+10 2.33433095e+08
1.40346290e+08 3.09357926e+08 0.00000000e+00 0.00000000e+00
0.00000000e+00 2.71028686e+10 1.80567074e+09 3.45144823e+08
0.00000000e+00 9.15504165e+10 8.93391307e+09 2.57731063e+10
1.59182431e+08 1.13652793e+11 0.00000000e+00 1.06492299e+09
4.42476364e+08 6.34324331e+10 7.75920540e+10 0.00000000e+00
0.00000000e+00 1.21820400e+09 6.39510881e+10 6.82481121e+08
0.00000000e+00 2.17894100e+09 8.19573109e+09 4.24212886e+10
0.00000000e+00 9.51509440e+08 1.27595323e+10 0.00000000e+00
2.30872508e+11 3.26356605e+09 4.71633039e+11 0.00000000e+00
0.00000000e+00 8.87837538e+09 0.00000000e+00 0.00000000e+00
0.00000000e+00 0.00000000e+00]

```

solution error = 369877.1213378906

6.783182016079081e-09 369877.1213378906

Out strength sequence [2.28286285e+08 1.15021444e+10 3.60605442e+09

5.11013530e+07

```

2.06461985e+10 6.83194940e+07 4.48927398e+10 4.73305319e+10
1.96182805e+08 6.41853983e+08 1.61414449e+09 3.58627275e+09
1.58358662e+08 1.56024136e+09 1.34224920e+11 2.14741612e+08
3.64060190e+08 4.56423870e+07 9.56462430e+08 4.59820258e+10
2.51547038e+09 3.83698059e+09 1.75199842e+08 2.32484601e+08
3.31241913e+08 2.20843126e+09 1.07089966e+11 1.07046100e+07
1.75409419e+08 2.16935186e+08 1.53997963e+10 1.76411982e+11
1.06414835e+10 1.30697110e+07 1.25212242e+09 3.74595755e+09
4.06737251e+09 3.42564373e+09 9.16062518e+08 1.50882467e+09
3.98195035e+10 4.61570220e+07 9.32752870e+07 3.98926508e+09
5.21348247e+09 6.22505716e+09 1.69773919e+09 1.08482097e+08
5.80453594e+08 3.38210753e+10 4.59946361e+08 2.34893813e+11
2.70071306e+09 1.78255318e+08 3.77254111e+11 1.45630451e+09
9.12266503e+09 2.40220260e+07 3.28050479e+09 7.39069488e+08
1.09932413e+08 4.92812881e+08 1.88851612e+08 2.37291984e+09
5.12678540e+10 1.26291717e+10 1.91703279e+09 3.13339217e+10
3.94245411e+10 1.61709521e+10 4.04809793e+10 1.69510053e+10
1.94112302e+11 1.79679997e+09 2.51815281e+11 7.40878429e+08
1.49357080e+09 1.62755892e+09 5.15631500e+06 8.88861844e+10
1.01879840e+10 2.51773794e+08 2.43735024e+08 2.11051022e+09
5.27786287e+08 9.12184707e+09 1.76704492e+09 6.37691971e+08
4.10100275e+08 6.04314294e+10 5.17490780e+07 4.47649496e+08
6.32081181e+08 1.59215901e+09 4.81802183e+10 3.55645762e+08
3.30555665e+08 6.96339512e+09 2.92596275e+08 1.29151613e+09
3.65680868e+08 1.55912961e+11 1.28177861e+10 5.33954319e+08

```

1.83713198e+08 8.75006399e+09 4.08164791e+10 4.86279761e+09
 9.75910071e+09 3.07961290e+09 2.69540554e+09 1.55928683e+09
 5.07849870e+09 1.86518806e+10 1.92122679e+10 2.14498745e+10
 3.41774008e+09 6.88279965e+09 5.89310876e+10 4.71241400e+07
 4.14689630e+07 1.28588579e+08 1.05824769e+08 5.87862680e+07
 6.81892000e+06 3.05947877e+10 5.50476555e+08 4.23532490e+07
 1.85611356e+08 6.04319739e+10 7.34023018e+09 2.08231423e+10
 1.96976262e+08 8.13913446e+10 3.46175903e+09 6.74853586e+08
 5.10063250e+08 7.20450562e+10 8.63643998e+10 3.41174501e+09
 3.81857518e+08 7.36810102e+08 4.95482165e+10 2.86104724e+08
 2.04114350e+07 1.83324643e+09 5.14281516e+09 1.83041834e+10
 9.68638475e+08 6.43136564e+08 6.21091412e+09 1.09592882e+10
 2.10439824e+11 2.46809486e+09 4.51961252e+11 2.68708953e+09
 2.89635990e+07 9.08084177e+09 5.42889784e+09 1.86454739e+09
 1.11470962e+09 1.74159112e+09]

In strength sequence [0.00000000e+00 1.06921382e+10 0.00000000e+00
 0.00000000e+00

1.96183801e+10 0.00000000e+00 5.44047703e+10 5.87201348e+10
 0.00000000e+00 0.00000000e+00 0.00000000e+00 5.35569277e+09
 0.00000000e+00 0.00000000e+00 1.41731301e+11 2.39477565e+08
 0.00000000e+00 0.00000000e+00 1.38924683e+09 5.32025281e+10
 0.00000000e+00 0.00000000e+00 4.63272719e+08 2.68848208e+08
 0.00000000e+00 1.04603635e+09 1.12371919e+11 0.00000000e+00
 2.03429028e+08 2.06036135e+08 1.46561925e+10 1.08534060e+11
 1.36973506e+10 6.24452170e+07 3.66419363e+08 3.16692973e+09
 2.44670377e+09 7.23293318e+09 3.52833905e+09 0.00000000e+00
 3.97151402e+10 0.00000000e+00 1.14379371e+08 0.00000000e+00
 4.03459295e+09 1.09432103e+10 2.53141897e+09 0.00000000e+00
 0.00000000e+00 2.82068884e+10 1.05476546e+09 2.37121334e+11
 0.00000000e+00 2.19487625e+08 3.83525323e+11 0.00000000e+00
 2.57913479e+10 1.27496171e+08 3.15059571e+09 8.17701283e+08
 0.00000000e+00 0.00000000e+00 0.00000000e+00 1.63445466e+09
 1.39849308e+11 1.45177330e+10 1.71615628e+09 3.25928825e+10
 3.94644085e+10 0.00000000e+00 3.06241706e+10 2.61099100e+10
 1.85996164e+11 0.00000000e+00 2.62670241e+11 3.16797989e+09
 3.77091353e+09 0.00000000e+00 3.35026990e+07 1.07217562e+11
 0.00000000e+00 4.95768779e+08 0.00000000e+00 1.70924498e+09
 0.00000000e+00 0.00000000e+00 3.47888181e+09 5.28823456e+08
 5.08840563e+08 5.63948879e+10 2.75181781e+08 0.00000000e+00
 0.00000000e+00 2.18977466e+09 5.45079730e+10 8.35891387e+08
 0.00000000e+00 8.69120249e+09 6.67903000e+08 0.00000000e+00
 0.00000000e+00 1.46928741e+11 1.35537625e+10 9.82654618e+08
 3.13414888e+08 0.00000000e+00 3.23350858e+10 4.25043318e+09
 0.00000000e+00 2.08790467e+09 0.00000000e+00 2.90104504e+09
 7.46709662e+09 0.00000000e+00 2.72730550e+10 3.40344861e+10
 0.00000000e+00 1.01674380e+10 0.00000000e+00 0.00000000e+00
 1.27086917e+08 2.95191113e+08 1.31062281e+08 0.00000000e+00
 0.00000000e+00 2.74621830e+10 0.00000000e+00 2.52796736e+08

0.00000000e+00 8.94129235e+10 9.06658541e+09 2.54545436e+10
0.00000000e+00 1.04007261e+11 0.00000000e+00 1.16942644e+09
5.61693182e+08 6.07375420e+10 7.52702601e+10 0.00000000e+00
0.00000000e+00 1.63698296e+09 5.86907180e+10 5.55473021e+08
0.00000000e+00 1.69661751e+09 8.39785024e+09 3.54627266e+10
0.00000000e+00 1.02897908e+09 0.00000000e+00 0.00000000e+00
2.15631410e+11 2.81010800e+09 4.51929765e+11 0.00000000e+00
0.00000000e+00 1.07294372e+10 0.00000000e+00 0.00000000e+00
7.11409091e+08 2.48712421e+09]

solution error = 240194.33235913515

2.3889619171768572e-09 240194.33235913515

Out strength sequence [2.64727195e+08 1.14239742e+10 4.47873741e+09

4.19394070e+07

2.29368465e+10 1.76746159e+08 4.77902669e+10 5.26312757e+10
8.00875108e+08 7.04883432e+08 2.23865502e+09 4.90409887e+09
2.31697898e+08 4.95513056e+09 4.09431224e+10 3.60726468e+08
4.20669079e+08 2.10765540e+07 8.43002899e+08 4.96870185e+10
2.33011205e+09 3.79045468e+09 2.87759835e+08 6.49508570e+07
1.17255422e+09 2.00466128e+09 1.09155817e+11 2.18698960e+07
2.20318140e+08 1.81116770e+08 1.55531289e+10 2.17053792e+11
1.26006889e+10 2.21693490e+07 1.26387762e+09 7.30080398e+09
4.52289333e+09 2.95512310e+09 9.25919167e+08 1.19288320e+09
4.02502019e+10 1.83934460e+07 1.09213796e+08 4.84682892e+09
5.56735691e+09 5.56697318e+09 2.63077980e+09 6.89261897e+08
6.12489888e+08 3.80850794e+10 3.62280733e+08 2.27781826e+11
2.66398310e+09 9.45436700e+07 3.68465045e+11 1.64227565e+09
9.72903587e+09 4.98143050e+07 4.32616093e+09 8.52650220e+08
6.92797770e+07 5.67041938e+08 3.45543837e+08 3.70604026e+09
5.19692526e+10 2.40286107e+10 2.15483504e+09 3.79445383e+10
4.75800193e+10 1.57964442e+10 5.70286526e+10 1.93529811e+10
1.94260426e+11 1.67641100e+09 2.56519926e+11 9.72274986e+08
5.31240361e+09 1.67393094e+09 7.50433700e+06 1.16308894e+11
1.22520814e+10 3.75644787e+08 2.27869376e+08 2.47645354e+09
6.28358141e+08 8.63849009e+09 2.87847676e+09 8.06573543e+08
4.97146075e+08 6.62473934e+10 1.16396275e+08 4.44462854e+08
6.39045382e+08 1.62035951e+09 6.31528106e+10 6.97034059e+08
3.65875749e+08 7.88356849e+09 3.20128707e+08 1.25781876e+09
5.78256077e+08 1.42336461e+11 1.26010678e+10 8.36869728e+08
3.02485638e+08 1.01964542e+10 4.10721225e+10 5.39068979e+09
9.00381820e+09 2.36647617e+09 2.09105712e+09 1.06610372e+09
5.52928464e+09 3.03219433e+10 2.36642522e+10 2.26291174e+10
6.36126538e+09 8.82868683e+09 6.79079663e+10 4.46174300e+07
5.00876090e+07 1.03724430e+08 1.79207404e+08 5.45555520e+07
1.66562750e+07 3.37490281e+10 6.83487809e+08 1.52356025e+08
8.90368330e+07 6.54038734e+10 7.58712675e+09 2.83540494e+10
1.61719925e+08 9.64544274e+10 4.19704436e+09 7.03504555e+08
6.07307711e+08 7.71397021e+10 8.35087527e+10 3.84050782e+09

3.41897912e+08 6.20241702e+08 5.82126787e+10 3.51624234e+08
 1.98887500e+07 2.96253459e+09 5.67964578e+09 2.52744260e+10
 1.16769358e+09 4.47649820e+08 1.01389012e+10 1.51406382e+10
 2.12146268e+11 2.56193583e+09 4.76900647e+11 2.60048449e+09
 5.15382620e+07 1.09279000e+10 1.05634260e+10 2.31906712e+09
 6.81678782e+08 2.00513606e+09]
 In strength sequence [1.13467064e+09 8.95813924e+09 0.00000000e+00
 3.33588145e+08
 2.49535143e+10 7.82003310e+08 5.77214007e+10 5.75583057e+10
 1.01195014e+09 1.70695279e+09 0.00000000e+00 0.00000000e+00
 1.06040168e+09 6.55198359e+09 2.34429125e+10 2.82674223e+08
 7.16716112e+08 1.83212315e+08 1.79343036e+09 5.08686747e+10
 0.00000000e+00 5.21533028e+09 7.72122205e+08 1.27134467e+08
 0.00000000e+00 1.29251263e+09 1.30996037e+11 2.54641287e+08
 7.90485470e+07 0.00000000e+00 1.36031894e+10 1.25639886e+11
 1.05578148e+10 7.95953280e+07 0.00000000e+00 5.91271017e+09
 2.88358236e+09 7.25498325e+09 3.43752713e+09 0.00000000e+00
 4.11683318e+10 0.00000000e+00 1.30066744e+08 0.00000000e+00
 2.99672698e+09 1.44542314e+10 3.04779545e+09 0.00000000e+00
 0.00000000e+00 2.91608347e+10 1.36244415e+09 2.23416442e+11
 8.95078653e+08 1.87557324e+08 3.57389976e+11 2.76101368e+09
 2.92869700e+10 2.01082267e+08 4.43884707e+09 5.39390202e+08
 0.00000000e+00 4.71121933e+08 0.00000000e+00 2.61082887e+09
 1.27659033e+11 2.61740500e+10 2.24007916e+09 4.94606325e+10
 2.36492493e+10 1.18865425e+10 4.38715014e+10 2.81159760e+10
 1.80191431e+11 2.75930669e+09 2.36150012e+11 3.14701775e+09
 3.57024001e+09 2.72633795e+09 4.06005300e+07 1.00216265e+11
 0.00000000e+00 0.00000000e+00 0.00000000e+00 2.69578252e+09
 6.00917597e+09 0.00000000e+00 4.51865217e+09 4.66047342e+08
 6.18371739e+08 4.83865349e+10 4.14908884e+08 8.10179476e+08
 0.00000000e+00 2.46683517e+09 8.92972018e+10 5.73296152e+08
 5.14900060e+08 1.08561484e+10 0.00000000e+00 0.00000000e+00
 1.41957583e+09 1.43390646e+11 1.39012482e+10 1.70721757e+09
 3.93156329e+08 4.35020218e+09 3.29111095e+10 4.66834869e+09
 0.00000000e+00 3.00634806e+09 0.00000000e+00 1.76170036e+09
 6.79268819e+09 3.09750237e+10 4.29906180e+10 3.88770484e+10
 0.00000000e+00 1.00830870e+10 2.91990399e+10 2.43132534e+08
 1.50670540e+08 3.51468690e+08 1.98800534e+08 0.00000000e+00
 3.74276360e+07 2.73178598e+10 1.60798314e+09 0.00000000e+00
 0.00000000e+00 8.42077257e+10 9.44956075e+09 2.30648941e+10
 0.00000000e+00 1.31823095e+11 5.31457836e+09 8.96017391e+08
 4.81536000e+08 6.08980916e+10 7.23810740e+10 0.00000000e+00
 0.00000000e+00 1.57018340e+09 4.72060478e+10 6.22386539e+08
 0.00000000e+00 2.71017121e+09 8.67452813e+09 3.94169346e+10
 1.33409295e+09 1.00709267e+09 1.14272187e+10 0.00000000e+00
 2.40778710e+11 3.30308073e+09 5.60811724e+11 0.00000000e+00
 0.00000000e+00 1.32149264e+10 0.00000000e+00 0.00000000e+00
 8.02845574e+08 0.00000000e+00]

```

solution error = 181065.18603515625
1.485290113123483e-09 181065.18603515625
Out strength sequence [3.56741139e+08 1.73315230e+10 6.30263181e+09
2.13795980e+08
2.73636268e+10 2.26486038e+08 5.94365394e+10 5.49611344e+10
1.32669827e+09 1.01881888e+09 2.74833342e+09 6.08779307e+09
3.64968605e+08 5.95148230e+09 2.63689304e+10 2.65874587e+08
6.57397009e+08 3.37129410e+07 1.11926409e+09 6.02636651e+10
3.62788801e+09 4.94713439e+09 2.55758994e+08 3.34235490e+07
1.70482641e+09 2.11679724e+09 1.11055709e+11 1.87291950e+07
9.35365900e+07 1.38857759e+08 1.82928931e+10 2.67117682e+11
1.34864883e+10 4.32679010e+07 1.66461740e+09 6.22374406e+09
3.70731349e+09 3.07981207e+09 1.03553690e+09 5.54187803e+08
4.24061397e+10 6.21170640e+07 7.87365760e+07 4.84376704e+09
6.02829739e+09 5.70548846e+09 3.02731634e+09 1.78543705e+09
5.14128708e+08 4.07000528e+10 2.76162960e+08 2.23947183e+11
3.04448384e+09 3.07976540e+07 3.79432472e+11 1.41966247e+09
9.74643619e+09 6.39140650e+07 4.80298889e+09 9.79953316e+08
8.47718240e+07 5.99376786e+08 3.04821909e+08 4.14141020e+09
5.26985107e+10 2.83154812e+10 2.28409609e+09 4.41556407e+10
5.48290949e+10 1.56887359e+10 6.69716869e+10 2.23778517e+10
1.93826708e+11 1.60531736e+09 2.59234313e+11 1.23094796e+09
7.52087426e+09 1.78566330e+09 3.46278700e+07 1.22264773e+11
1.04993160e+10 3.12961470e+08 2.38760866e+08 3.21056879e+09
8.10666717e+08 7.92055290e+09 3.94418054e+09 1.05849433e+09
4.67960526e+08 7.53514711e+10 1.91645652e+08 2.85511817e+08
5.16689927e+08 1.65163716e+09 6.37503049e+10 8.51701434e+08
4.96587927e+08 8.06276520e+09 5.55968234e+08 2.80658294e+09
7.44553700e+08 1.38981679e+11 1.42815135e+10 1.12558332e+09
1.94716412e+08 1.08990372e+10 4.24752561e+10 9.85694667e+09
1.09963867e+10 2.70266045e+09 1.68671682e+09 1.32602981e+09
6.18870900e+09 3.39137708e+10 3.03580800e+10 2.20389929e+10
7.29331615e+09 1.17077770e+10 8.97872730e+10 7.09841100e+07
6.24396060e+07 7.18574650e+07 1.47436616e+08 6.24748930e+07
1.33021380e+07 4.13646306e+10 7.44352082e+08 2.54835464e+08
9.66231700e+07 7.43382459e+10 8.15071821e+09 2.26753191e+10
8.45651890e+07 9.70533281e+10 4.49094142e+09 1.82465785e+09
5.30460177e+08 7.20294478e+10 8.91170162e+10 6.06871802e+09
4.20726179e+08 6.24194633e+08 6.50437722e+10 4.08584382e+08
2.58088740e+07 4.23634325e+09 6.16842470e+09 2.88940515e+10
2.43504433e+09 3.63235480e+08 1.39725115e+10 2.12653426e+10
2.16746704e+11 2.33862033e+09 5.01376962e+11 2.71073312e+09
3.74065130e+07 1.30713515e+10 1.36945006e+10 2.96707971e+09
6.44842040e+08 1.77007435e+09]
In strength sequence [1.31513335e+09 9.69463656e+09 0.00000000e+00
0.00000000e+00
1.95308475e+10 0.00000000e+00 5.78097256e+10 5.70295537e+10

```

```

1.40588453e+09 1.85818671e+09 4.34654105e+09 8.44962776e+09
1.05806324e+09 8.06650432e+09 2.20388176e+10 4.56703314e+08
5.96017844e+08 0.00000000e+00 1.70203364e+09 5.70187399e+10
1.15205897e+09 7.10667566e+09 5.39332627e+08 1.35698911e+08
1.33337725e+09 1.78397098e+09 1.39965316e+11 2.39040339e+08
5.89145300e+07 0.00000000e+00 1.57811135e+10 1.68991620e+11
1.25191303e+10 0.00000000e+00 0.00000000e+00 6.22738079e+09
0.00000000e+00 8.39762409e+09 3.66828659e+09 0.00000000e+00
4.12915339e+10 0.00000000e+00 1.28552416e+08 5.24521700e+09
5.28679811e+09 1.14065521e+10 3.80092257e+09 0.00000000e+00
0.00000000e+00 2.90975198e+10 1.77824345e+09 2.22289111e+11
9.95826045e+08 1.35684663e+08 3.58924035e+11 0.00000000e+00
2.68210924e+10 2.10447243e+08 5.45995574e+09 5.73347117e+08
0.00000000e+00 5.11169888e+08 0.00000000e+00 2.51517258e+09
1.39382343e+11 3.12121990e+10 2.00562969e+09 3.73527393e+10
3.04560726e+10 1.54594371e+10 4.36408911e+10 2.91775640e+10
1.87882181e+11 3.27069156e+09 2.52626339e+11 3.88451169e+09
6.24082850e+09 3.76901475e+09 0.00000000e+00 1.15431115e+11
7.73052658e+09 0.00000000e+00 0.00000000e+00 3.15221654e+09
7.10197963e+09 0.00000000e+00 5.97302919e+09 7.65217008e+08
5.09131152e+08 5.96344965e+10 3.98751132e+08 9.90310914e+08
0.00000000e+00 2.14039680e+09 1.13977600e+11 8.67120098e+08
6.37258187e+08 1.07522709e+10 3.69048623e+08 0.00000000e+00
0.00000000e+00 1.42169268e+11 1.29005590e+10 1.71463027e+09
3.23215521e+08 7.54785364e+09 3.16522189e+10 5.78198387e+09
0.00000000e+00 2.52488568e+09 1.30263715e+09 2.16167735e+09
7.23133377e+09 2.98049990e+10 4.68140160e+10 3.86281538e+10
3.71759933e+09 1.48283680e+10 3.97987638e+10 2.69187920e+08
1.85954789e+08 2.73022117e+08 1.84484275e+08 1.35392054e+08
2.81187290e+07 3.04322928e+10 1.72965593e+09 3.05541453e+08
0.00000000e+00 8.42168525e+10 9.46142152e+09 9.36203718e+09
0.00000000e+00 1.38664489e+11 5.41726965e+09 1.94256262e+09
3.99797000e+08 5.99785088e+10 8.05559331e+10 0.00000000e+00
0.00000000e+00 1.63923138e+09 5.76977281e+10 3.40497764e+08
0.00000000e+00 3.88816185e+09 9.97268758e+09 4.05078339e+10
0.00000000e+00 9.70804655e+08 1.52111145e+10 0.00000000e+00
2.54621298e+11 3.02462586e+09 5.86248428e+11 0.00000000e+00
0.00000000e+00 1.58698310e+10 0.00000000e+00 0.00000000e+00
1.28376516e+09 1.20051569e+09]

```

```

solution error = 435455.66259765625
3.313960661444071e-09 435455.66259765625

```

```
[179]: nemtropy_directed_results
```

```
[179]: {'WDN_1997': {'nemtropy_graph': <NEMtropy.graph_classes.DirectedGraph at 0x13972b690>,

```

```

'fit': None,
'error_degree': 3.84389409191499e-09,
'error_strength': 73029.06451416016},
'WDN_1992': {'nentropy_graph': <NEMtropy.graph_classes.DirectedGraph at
0x1464d57d0>,
'fit': None,
'error_degree': 3.1084255169844255e-09,
'error_strength': 345585.3070678711},
'WDN_2000': {'nentropy_graph': <NEMtropy.graph_classes.DirectedGraph at
0x139718710>,
'fit': None,
'error_degree': 2.1081412171497504e-09,
'error_strength': 190784.88024902344},
'WDN_1998': {'nentropy_graph': <NEMtropy.graph_classes.DirectedGraph at
0x14384d010>,
'fit': None,
'error_degree': 1.9816468466160586e-09,
'error_strength': 225387.19116210938},
'WDN_1994': {'nentropy_graph': <NEMtropy.graph_classes.DirectedGraph at
0x1390e8b10>,
'fit': None,
'error_degree': 5.512987399924896e-09,
'error_strength': 206719.9835205078},
'WDN_1993': {'nentropy_graph': <NEMtropy.graph_classes.DirectedGraph at
0x1438301d0>,
'fit': None,
'error_degree': 6.645926475812303e-09,
'error_strength': 109617.70764160156},
'WDN_2002': {'nentropy_graph': <NEMtropy.graph_classes.DirectedGraph at
0x1439808d0>,
'fit': None,
'error_degree': 3.6602507691441133e-09,
'error_strength': 130727.1923828125},
'WDN_1996': {'nentropy_graph': <NEMtropy.graph_classes.DirectedGraph at
0x143388350>,
'fit': None,
'error_degree': 6.783182016079081e-09,
'error_strength': 369877.1213378906},
'WDN_1995': {'nentropy_graph': <NEMtropy.graph_classes.DirectedGraph at
0x13966c410>,
'fit': None,
'error_degree': 2.3889619171768572e-09,
'error_strength': 240194.33235913515},
'WDN_1999': {'nentropy_graph': <NEMtropy.graph_classes.DirectedGraph at
0x13966be10>,
'fit': None,
'error_degree': 1.485290113123483e-09,

```

```

    'error_strength': 181065.18603515625},
    'WDN_2001': {'nentropy_graph': <NEMtropy.graph_classes.DirectedGraph at
0x143770110>,
    'fit': None,
    'error_degree': 3.313960661444071e-09,
    'error_strength': 435455.66259765625}}

```

```

[180]: for graph_name, graph in graphs_wtn_undirected.items():

    # Plot the networkx graph
    # plot_networkx(graph=graph, graph_name=graph_name, is_directed=False)

    # Extract necessary information from the NetworkX graph
    adj_weigh, edges, degrees, strengths = to_entropy_graph(graph)

    # Instantiate DirectedGraph with the extracted information
    # TODO: Double check, if we might need UndirectedGraph
    nentropy_graph = UndirectedGraph(
        adjacency=adj_weigh,
        edgelist=edges,
        degree_sequence=np.array(list(degrees.values())),
        strength_sequence=np.array(list(strengths.values())))
    )

    # Plot the entropy graph
    # plot_entropy_graph(adj_weigh=adj_weigh, graph_name=graph_name,
↪ is_directed=False)
    # We can see the graph sequence of the graph by tapping

    # Create the fit
    fit = nentropy_graph.solve_tool(model="cm_exp",
                                    method="newton",
                                    initial_guess="random",
                                    adjacency="dcm_exp",
                                    method_adjacency="newton",
                                    max_steps=MAX_STEPS)

    nentropy_undirected_results[graph_name] = {'nentropy_graph': nentropy_graph,
↪ 'fit': fit, 'error_degree': nentropy_graph.error_degree, 'error_strength':
↪ nentropy_graph.error_strength}
    print(nentropy_graph.error_degree, nentropy_graph.error_strength)

```

```

solution error = 2.2947688194108196e-09
2.2947688194108196e-09 None

```

```

solution error = 6.044842848496046e-09

```

```

6.044842848496046e-09 None

solution error = 2.2773178898205515e-09
2.2773178898205515e-09 None

solution error = 1.6948717984632822e-09
1.6948717984632822e-09 None

solution error = 2.8035458399244817e-09
2.8035458399244817e-09 None

solution error = 7.512738875448122e-09
7.512738875448122e-09 None

solution error = 2.058698100881884e-09
2.058698100881884e-09 None

solution error = 1.957971562660532e-09
1.957971562660532e-09 None

solution error = 2.09220729630033e-09
2.09220729630033e-09 None

solution error = 8.603535661677597e-10
8.603535661677597e-10 None

solution error = 8.608907364759943e-09
8.608907364759943e-09 None

```

```
[181]: nemtropy_undirected_results
```

```

[181]: {'WDN_1997': {'nemtropy_graph': <NEMtropy.graph_classes.UndirectedGraph at
0x1461904d0>,
  'fit': None,
  'error_degree': 2.2947688194108196e-09,
  'error_strength': None},
  'WDN_1992': {'nemtropy_graph': <NEMtropy.graph_classes.UndirectedGraph at
0x1313f6c10>,
  'fit': None,
  'error_degree': 6.044842848496046e-09,
  'error_strength': None},
  'WDN_2000': {'nemtropy_graph': <NEMtropy.graph_classes.UndirectedGraph at
0x1461919d0>,
  'fit': None,
  'error_degree': 2.2773178898205515e-09,
  'error_strength': None},
  'WDN_1998': {'nemtropy_graph': <NEMtropy.graph_classes.UndirectedGraph at

```

```

0x1439a6e50>,
  'fit': None,
  'error_degree': 1.6948717984632822e-09,
  'error_strength': None},
'WDN_1994': {'nentropy_graph': <NEMtropy.graph_classes.UndirectedGraph at
0x1434ac990>,
  'fit': None,
  'error_degree': 2.8035458399244817e-09,
  'error_strength': None},
'WDN_1993': {'nentropy_graph': <NEMtropy.graph_classes.UndirectedGraph at
0x13189f410>,
  'fit': None,
  'error_degree': 7.512738875448122e-09,
  'error_strength': None},
'WDN_2002': {'nentropy_graph': <NEMtropy.graph_classes.UndirectedGraph at
0x143838150>,
  'fit': None,
  'error_degree': 2.058698100881884e-09,
  'error_strength': None},
'WDN_1996': {'nentropy_graph': <NEMtropy.graph_classes.UndirectedGraph at
0x1395ac810>,
  'fit': None,
  'error_degree': 1.957971562660532e-09,
  'error_strength': None},
'WDN_1995': {'nentropy_graph': <NEMtropy.graph_classes.UndirectedGraph at
0x1438f8350>,
  'fit': None,
  'error_degree': 2.09220729630033e-09,
  'error_strength': None},
'WDN_1999': {'nentropy_graph': <NEMtropy.graph_classes.UndirectedGraph at
0x14346cdd0>,
  'fit': None,
  'error_degree': 8.603535661677597e-10,
  'error_strength': None},
'WDN_2001': {'nentropy_graph': <NEMtropy.graph_classes.UndirectedGraph at
0x143733850>,
  'fit': None,
  'error_degree': 8.608907364759943e-09,
  'error_strength': None}}

```

2.3 3. Sample 30 networks from the obtained ERGM distributions and measure the strength assortativity on all of them. Calculate average and standard deviation of each assortativity typology.

```
[182]: SAMPLE_SIZE = 30
```

```

[183]: for graph_name, results in nemtropy_directed_results.items():
    graph = results['nemtropy_graph']
    error_degree = results['error_degree']
    error_strength = results['error_strength']

    # Generate 30 ensembles copies using ensemble_sampler
    graph.ensemble_sampler(SAMPLE_SIZE, cpu_n=2,
    ↪output_dir=f"{OUTPUT_PATH_ERGMS}/{graph_name}-directed-sample/")

    sampled_graphs = []
    assortativity_values = []
    for i in range(0, SAMPLE_SIZE, 1):
        # We read the edgelist
        edgelist_ens = np.loadtxt(f"{OUTPUT_PATH_ERGMS}/{
    ↪graph_name}-directed-sample/{i}.txt")

        # and build the adjacency matrix
        ens_adj = build_adjacency_from_edgelist(edgelist = edgelist_ens,
                                                is_directed = True,
                                                is_sparse = False,
                                                is_weighted = True)

        sampled_graph = to_networkx_graph(ens_adj)
        sampled_graphs.append(sampled_graph)

        # measure strength assortativity
        assortativities_sampled = calculate_assortativity(sampled_graph)
        assortativity_values.append(assortativities_sampled)

    categories = list(assortativity_values[0].keys())

    # Initialize dictionaries to store mean and standard deviation for each
    ↪category
    mean_values = {}
    std_values = {}

    # Loop through each category
    for category in categories:
        # Extract values for the current category across all dictionaries
        category_values = [data[category] for data in assortativity_values]

        # Calculate mean and standard
        mean = np.mean(category_values)
        std = np.std(category_values)

        # Store mean and standard deviation for the category
        mean_values[category] = mean

```

```

std_values[category] = std

results['samples'] = sampled_graphs
results['assortativity_means'] = mean_values
results['assortativity_stds'] = std_values
results['assortativity_data'] = assortativities[graph_name]

```

```
[184]: nemtropy_directed_results
```

```

[184]: {'WDN_1997': {'nemtropy_graph': <NEMtropy.graph_classes.DirectedGraph at
0x13972b690>,
'fit': None,
'error_degree': 3.84389409191499e-09,
'error_strength': 73029.06451416016,
'samples': [<networkx.classes.digraph.DiGraph at 0x13958fe90>,
<networkx.classes.digraph.DiGraph at 0x144402c10>,
<networkx.classes.digraph.DiGraph at 0x143fcc7d0>,
<networkx.classes.digraph.DiGraph at 0x1443bfc50>,
<networkx.classes.digraph.DiGraph at 0x144241a90>,
<networkx.classes.digraph.DiGraph at 0x1446bc8d0>,
<networkx.classes.digraph.DiGraph at 0x14529ac50>,
<networkx.classes.digraph.DiGraph at 0x1443bee90>,
<networkx.classes.digraph.DiGraph at 0x14557c050>,
<networkx.classes.digraph.DiGraph at 0x1478b4510>,
<networkx.classes.digraph.DiGraph at 0x139717ad0>,
<networkx.classes.digraph.DiGraph at 0x1450c9210>,
<networkx.classes.digraph.DiGraph at 0x142bd7590>,
<networkx.classes.digraph.DiGraph at 0x148355ad0>,
<networkx.classes.digraph.DiGraph at 0x1444be750>,
<networkx.classes.digraph.DiGraph at 0x148c506d0>,
<networkx.classes.digraph.DiGraph at 0x148e219d0>,
<networkx.classes.digraph.DiGraph at 0x148e49610>,
<networkx.classes.digraph.DiGraph at 0x1443bfd90>,
<networkx.classes.digraph.DiGraph at 0x143863d90>,
<networkx.classes.digraph.DiGraph at 0x14997bd50>,
<networkx.classes.digraph.DiGraph at 0x149f07810>,
<networkx.classes.digraph.DiGraph at 0x149f07790>,
<networkx.classes.digraph.DiGraph at 0x14a2b10d0>,
<networkx.classes.digraph.DiGraph at 0x143ad0450>,
<networkx.classes.digraph.DiGraph at 0x1497f08d0>,
<networkx.classes.digraph.DiGraph at 0x14abb05d0>,
<networkx.classes.digraph.DiGraph at 0x147d15c10>,
<networkx.classes.digraph.DiGraph at 0x14b3706d0>,
<networkx.classes.digraph.DiGraph at 0x14b3c45d0>],
'assortativity_means': {'In-In Assortativity': -0.06730087236716178,
'In-Out Assortativity': -0.06439285138092891,
'Out-In Assortativity': -0.06472194634943872,

```

```

'Out-Out Assortativity': -0.06435956667855361},
'assortativity_stds': {'In-In Assortativity': 0.013024494297945641,
'In-Out Assortativity': 0.012303869141582479,
'Out-In Assortativity': 0.012396954132747788,
'Out-Out Assortativity': 0.011949758528564784},
'assortativity_data': {'In-In Assortativity': -0.07368635207387701,
'In-Out Assortativity': -0.07132955497657792,
'Out-In Assortativity': -0.07171683897072376,
'Out-Out Assortativity': -0.06970079872009399}},
'WDN_1992': {'nentropy_graph': <NEMtropy.graph_classes.DirectedGraph at
0x1464d57d0>,
'fit': None,
'error_degree': 3.1084255169844255e-09,
'error_strength': 345585.3070678711,
'samples': [<networkx.classes.digraph.DiGraph at 0x143acbbd0>,
<networkx.classes.digraph.DiGraph at 0x14446c950>,
<networkx.classes.digraph.DiGraph at 0x14446f210>,
<networkx.classes.digraph.DiGraph at 0x13962aa10>,
<networkx.classes.digraph.DiGraph at 0x145cf7e50>,
<networkx.classes.digraph.DiGraph at 0x14bd1d2d0>,
<networkx.classes.digraph.DiGraph at 0x14bc8bd50>,
<networkx.classes.digraph.DiGraph at 0x14c0bc610>,
<networkx.classes.digraph.DiGraph at 0x14c1f3ad0>,
<networkx.classes.digraph.DiGraph at 0x14c0fcf90>,
<networkx.classes.digraph.DiGraph at 0x14c39ac10>,
<networkx.classes.digraph.DiGraph at 0x14bb5b150>,
<networkx.classes.digraph.DiGraph at 0x143840650>,
<networkx.classes.digraph.DiGraph at 0x142c67f50>,
<networkx.classes.digraph.DiGraph at 0x14c780590>,
<networkx.classes.digraph.DiGraph at 0x14cb16d10>,
<networkx.classes.digraph.DiGraph at 0x14cb690d0>,
<networkx.classes.digraph.DiGraph at 0x148355410>,
<networkx.classes.digraph.DiGraph at 0x14d08c6d0>,
<networkx.classes.digraph.DiGraph at 0x14bce2310>,
<networkx.classes.digraph.DiGraph at 0x14d1a7fd0>,
<networkx.classes.digraph.DiGraph at 0x14bc73cd0>,
<networkx.classes.digraph.DiGraph at 0x14d1cb0d0>,
<networkx.classes.digraph.DiGraph at 0x14d7546d0>,
<networkx.classes.digraph.DiGraph at 0x14d710790>,
<networkx.classes.digraph.DiGraph at 0x14d731a90>,
<networkx.classes.digraph.DiGraph at 0x14403ee90>,
<networkx.classes.digraph.DiGraph at 0x14dae7490>,
<networkx.classes.digraph.DiGraph at 0x14dc5d1d0>,
<networkx.classes.digraph.DiGraph at 0x14df05d90>],
'assortativity_means': {'In-In Assortativity': -0.04585973820700722,
'In-Out Assortativity': -0.04094018622057668,
'Out-In Assortativity': -0.052267734295828,

```

```

'Out-Out Assortativity': -0.05027649921820144},
'assortativity_stds': {'In-In Assortativity': 0.007078426505857442,
'In-Out Assortativity': 0.008686228701483964,
'Out-In Assortativity': 0.00983659881198889,
'Out-Out Assortativity': 0.009663710387968897},
'assortativity_data': {'In-In Assortativity': -0.04868776409103533,
'In-Out Assortativity': -0.046647404340923876,
'Out-In Assortativity': -0.059869793086991335,
'Out-Out Assortativity': -0.05760318920541264}},
'WDN_2000': {'nentropy_graph': <NEMtropy.graph_classes.DirectedGraph at
0x139718710>,
'fit': None,
'error_degree': 2.1081412171497504e-09,
'error_strength': 190784.88024902344,
'samples': [<networkx.classes.digraph.DiGraph at 0x144275d90>,
<networkx.classes.digraph.DiGraph at 0x14e6c8690>,
<networkx.classes.digraph.DiGraph at 0x14e997310>,
<networkx.classes.digraph.DiGraph at 0x14dee1fd0>,
<networkx.classes.digraph.DiGraph at 0x14ea10790>,
<networkx.classes.digraph.DiGraph at 0x14efb7590>,
<networkx.classes.digraph.DiGraph at 0x14e7bbc50>,
<networkx.classes.digraph.DiGraph at 0x14f3c46d0>,
<networkx.classes.digraph.DiGraph at 0x14f61c590>,
<networkx.classes.digraph.DiGraph at 0x14e7bbd90>,
<networkx.classes.digraph.DiGraph at 0x14fad65d0>,
<networkx.classes.digraph.DiGraph at 0x14fca0890>,
<networkx.classes.digraph.DiGraph at 0x14e9a1e50>,
<networkx.classes.digraph.DiGraph at 0x14ffaea90>,
<networkx.classes.digraph.DiGraph at 0x14f65ca50>,
<networkx.classes.digraph.DiGraph at 0x14dee4790>,
<networkx.classes.digraph.DiGraph at 0x1508a46d0>,
<networkx.classes.digraph.DiGraph at 0x15169c610>,
<networkx.classes.digraph.DiGraph at 0x1452cec90>,
<networkx.classes.digraph.DiGraph at 0x151c64610>,
<networkx.classes.digraph.DiGraph at 0x14e2e5350>,
<networkx.classes.digraph.DiGraph at 0x145cccb50>,
<networkx.classes.digraph.DiGraph at 0x15257c6d0>,
<networkx.classes.digraph.DiGraph at 0x152300850>,
<networkx.classes.digraph.DiGraph at 0x1527b1390>,
<networkx.classes.digraph.DiGraph at 0x148587590>,
<networkx.classes.digraph.DiGraph at 0x1530fbc90>,
<networkx.classes.digraph.DiGraph at 0x1534346d0>,
<networkx.classes.digraph.DiGraph at 0x15377c6d0>,
<networkx.classes.digraph.DiGraph at 0x153b78610>],
'assortativity_means': {'In-In Assortativity': -0.06076264070882446,
'In-Out Assortativity': -0.0607778526922694,
'Out-In Assortativity': -0.061521904201264105,

```

```

'Out-Out Assortativity': -0.06417720653648396},
'assortativity_stds': {'In-In Assortativity': 0.008589190266559146,
'In-Out Assortativity': 0.007420038599066644,
'Out-In Assortativity': 0.00750040915995674,
'Out-Out Assortativity': 0.008298817420865781},
'assortativity_data': {'In-In Assortativity': -0.0631970717497539,
'In-Out Assortativity': -0.06467170213846346,
'Out-In Assortativity': -0.06539073425903164,
'Out-Out Assortativity': -0.06730072607061888}},
'WDN_1998': {'nentropy_graph': <NEMtropy.graph_classes.DirectedGraph at
0x14384d010>,
'fit': None,
'error_degree': 1.9816468466160586e-09,
'error_strength': 225387.19116210938,
'samples': [<networkx.classes.digraph.DiGraph at 0x14b8ac0d0>,
<networkx.classes.digraph.DiGraph at 0x1542ba390>,
<networkx.classes.digraph.DiGraph at 0x1545bbf50>,
<networkx.classes.digraph.DiGraph at 0x154323e50>,
<networkx.classes.digraph.DiGraph at 0x154aac690>,
<networkx.classes.digraph.DiGraph at 0x153f32cd0>,
<networkx.classes.digraph.DiGraph at 0x154f3bfd0>,
<networkx.classes.digraph.DiGraph at 0x1545f7cd0>,
<networkx.classes.digraph.DiGraph at 0x1544f7e10>,
<networkx.classes.digraph.DiGraph at 0x15443d650>,
<networkx.classes.digraph.DiGraph at 0x153b14150>,
<networkx.classes.digraph.DiGraph at 0x155678850>,
<networkx.classes.digraph.DiGraph at 0x155ed8a90>,
<networkx.classes.digraph.DiGraph at 0x15643c750>,
<networkx.classes.digraph.DiGraph at 0x1567ec750>,
<networkx.classes.digraph.DiGraph at 0x15677c690>,
<networkx.classes.digraph.DiGraph at 0x1567cc490>,
<networkx.classes.digraph.DiGraph at 0x156fb4750>,
<networkx.classes.digraph.DiGraph at 0x157400750>,
<networkx.classes.digraph.DiGraph at 0x1573eca90>,
<networkx.classes.digraph.DiGraph at 0x1544bb2d0>,
<networkx.classes.digraph.DiGraph at 0x15788c8d0>,
<networkx.classes.digraph.DiGraph at 0x1567a6010>,
<networkx.classes.digraph.DiGraph at 0x153f33d10>,
<networkx.classes.digraph.DiGraph at 0x1544f7fd0>,
<networkx.classes.digraph.DiGraph at 0x1545bbe10>,
<networkx.classes.digraph.DiGraph at 0x1584b2b10>,
<networkx.classes.digraph.DiGraph at 0x158263790>,
<networkx.classes.digraph.DiGraph at 0x158261e50>,
<networkx.classes.digraph.DiGraph at 0x1592f3e50>],
'assortativity_means': {'In-In Assortativity': -0.06169809101460549,
'In-Out Assortativity': -0.06001799139531442,
'Out-In Assortativity': -0.06032672273113892,

```

```

'Out-Out Assortativity': -0.060627397225747365},
'assortativity_stds': {'In-In Assortativity': 0.009897346842922688,
'In-Out Assortativity': 0.008691308304428711,
'Out-In Assortativity': 0.008857289680495992,
'Out-Out Assortativity': 0.008518573340157326},
'assortativity_data': {'In-In Assortativity': -0.0655705781839572,
'In-Out Assortativity': -0.06540616504184192,
'Out-In Assortativity': -0.06572057397949527,
'Out-Out Assortativity': -0.06583617855060356}},
'WDN_1994': {'nentropy_graph': <NEMtropy.graph_classes.DirectedGraph at
0x1390e8b10>,
'fit': None,
'error_degree': 5.512987399924896e-09,
'error_strength': 206719.9835205078,
'samples': [<networkx.classes.digraph.DiGraph at 0x1547081d0>,
<networkx.classes.digraph.DiGraph at 0x158c09090>,
<networkx.classes.digraph.DiGraph at 0x159659f90>,
<networkx.classes.digraph.DiGraph at 0x15990b650>,
<networkx.classes.digraph.DiGraph at 0x159637d50>,
<networkx.classes.digraph.DiGraph at 0x159e98bd0>,
<networkx.classes.digraph.DiGraph at 0x15a0b7610>,
<networkx.classes.digraph.DiGraph at 0x15a554750>,
<networkx.classes.digraph.DiGraph at 0x1593374d0>,
<networkx.classes.digraph.DiGraph at 0x159835450>,
<networkx.classes.digraph.DiGraph at 0x1596372d0>,
<networkx.classes.digraph.DiGraph at 0x153f36e90>,
<networkx.classes.digraph.DiGraph at 0x1558ffa50>,
<networkx.classes.digraph.DiGraph at 0x15af11b10>,
<networkx.classes.digraph.DiGraph at 0x15998b2d0>,
<networkx.classes.digraph.DiGraph at 0x159635a90>,
<networkx.classes.digraph.DiGraph at 0x159bac850>,
<networkx.classes.digraph.DiGraph at 0x15b7e42d0>,
<networkx.classes.digraph.DiGraph at 0x14614c750>,
<networkx.classes.digraph.DiGraph at 0x15bb5b390>,
<networkx.classes.digraph.DiGraph at 0x159837d90>,
<networkx.classes.digraph.DiGraph at 0x15c03cad0>,
<networkx.classes.digraph.DiGraph at 0x159637e10>,
<networkx.classes.digraph.DiGraph at 0x15c6346d0>,
<networkx.classes.digraph.DiGraph at 0x15c6b0790>,
<networkx.classes.digraph.DiGraph at 0x15c8b7ed0>,
<networkx.classes.digraph.DiGraph at 0x15c85d410>,
<networkx.classes.digraph.DiGraph at 0x15c85c4d0>,
<networkx.classes.digraph.DiGraph at 0x15cd085d0>,
<networkx.classes.digraph.DiGraph at 0x15d061a90>],
'assortativity_means': {'In-In Assortativity': -0.07535259608884769,
'In-Out Assortativity': -0.07309343052811851,
'Out-In Assortativity': -0.0770859860851278,

```

```

'Out-Out Assortativity': -0.07763452109649437},
'assortativity_stds': {'In-In Assortativity': 0.010712215192268115,
'In-Out Assortativity': 0.01104786829164509,
'Out-In Assortativity': 0.011742189299834822,
'Out-Out Assortativity': 0.013379576121234533},
'assortativity_data': {'In-In Assortativity': -0.07868988970611883,
'In-Out Assortativity': -0.07622523439269807,
'Out-In Assortativity': -0.08022501617948047,
'Out-Out Assortativity': -0.07800143058821393}},
'WDN_1993': {'nentropy_graph': <NEMtropy.graph_classes.DirectedGraph at
0x1438301d0>,
'fit': None,
'error_degree': 6.645926475812303e-09,
'error_strength': 109617.70764160156,
'samples': [<networkx.classes.digraph.DiGraph at 0x159b90c50>,
<networkx.classes.digraph.DiGraph at 0x15a0e9250>,
<networkx.classes.digraph.DiGraph at 0x14add4f50>,
<networkx.classes.digraph.DiGraph at 0x15d746910>,
<networkx.classes.digraph.DiGraph at 0x15d5e5250>,
<networkx.classes.digraph.DiGraph at 0x15c270a50>,
<networkx.classes.digraph.DiGraph at 0x15db646d0>,
<networkx.classes.digraph.DiGraph at 0x15e060750>,
<networkx.classes.digraph.DiGraph at 0x15e060e90>,
<networkx.classes.digraph.DiGraph at 0x15e060690>,
<networkx.classes.digraph.DiGraph at 0x142c8bd50>,
<networkx.classes.digraph.DiGraph at 0x15e500350>,
<networkx.classes.digraph.DiGraph at 0x15e667e90>,
<networkx.classes.digraph.DiGraph at 0x15e7074d0>,
<networkx.classes.digraph.DiGraph at 0x15e9e3c90>,
<networkx.classes.digraph.DiGraph at 0x15ecfbb90>,
<networkx.classes.digraph.DiGraph at 0x15eb85750>,
<networkx.classes.digraph.DiGraph at 0x15d527d50>,
<networkx.classes.digraph.DiGraph at 0x15f088290>,
<networkx.classes.digraph.DiGraph at 0x15f242ed0>,
<networkx.classes.digraph.DiGraph at 0x15f279710>,
<networkx.classes.digraph.DiGraph at 0x15f53f990>,
<networkx.classes.digraph.DiGraph at 0x15d667290>,
<networkx.classes.digraph.DiGraph at 0x15f77b7d0>,
<networkx.classes.digraph.DiGraph at 0x15fc5c0d0>,
<networkx.classes.digraph.DiGraph at 0x15fc74a90>,
<networkx.classes.digraph.DiGraph at 0x15e014850>,
<networkx.classes.digraph.DiGraph at 0x146179b50>,
<networkx.classes.digraph.DiGraph at 0x1464ff150>,
<networkx.classes.digraph.DiGraph at 0x1602d6b50>],
'assortativity_means': {'In-In Assortativity': -0.06019613986241048,
'In-Out Assortativity': -0.055263894006916954,
'Out-In Assortativity': -0.06413226937301335,

```

```

'Out-Out Assortativity': -0.06325466621744387},
'assortativity_stds': {'In-In Assortativity': 0.012290290217363931,
'In-Out Assortativity': 0.012671043032476237,
'Out-In Assortativity': 0.014865941178408677,
'Out-Out Assortativity': 0.01438754221217995},
'assortativity_data': {'In-In Assortativity': -0.06537929984057925,
'In-Out Assortativity': -0.062189867187614475,
'Out-In Assortativity': -0.07202068872957715,
'Out-Out Assortativity': -0.06910618411247677}},
'WDN_2002': {'nentropy_graph': <NEMtropy.graph_classes.DirectedGraph at
0x1439808d0>,
'fit': None,
'error_degree': 3.6602507691441133e-09,
'error_strength': 130727.1923828125,
'samples': [<networkx.classes.digraph.DiGraph at 0x15d8ea610>,
<networkx.classes.digraph.DiGraph at 0x15680acd0>,
<networkx.classes.digraph.DiGraph at 0x160d94d10>,
<networkx.classes.digraph.DiGraph at 0x160e3c510>,
<networkx.classes.digraph.DiGraph at 0x15fc6a3d0>,
<networkx.classes.digraph.DiGraph at 0x159751e90>,
<networkx.classes.digraph.DiGraph at 0x161369150>,
<networkx.classes.digraph.DiGraph at 0x1615949d0>,
<networkx.classes.digraph.DiGraph at 0x161ceb2d0>,
<networkx.classes.digraph.DiGraph at 0x161f300d0>,
<networkx.classes.digraph.DiGraph at 0x160b02910>,
<networkx.classes.digraph.DiGraph at 0x1605afd90>,
<networkx.classes.digraph.DiGraph at 0x1628e49d0>,
<networkx.classes.digraph.DiGraph at 0x160c347d0>,
<networkx.classes.digraph.DiGraph at 0x160ae32d0>,
<networkx.classes.digraph.DiGraph at 0x15d8c3c10>,
<networkx.classes.digraph.DiGraph at 0x163714750>,
<networkx.classes.digraph.DiGraph at 0x162b4be90>,
<networkx.classes.digraph.DiGraph at 0x163505f90>,
<networkx.classes.digraph.DiGraph at 0x164104750>,
<networkx.classes.digraph.DiGraph at 0x160ea8b90>,
<networkx.classes.digraph.DiGraph at 0x1647c4690>,
<networkx.classes.digraph.DiGraph at 0x1646745d0>,
<networkx.classes.digraph.DiGraph at 0x1647f7710>,
<networkx.classes.digraph.DiGraph at 0x164316150>,
<networkx.classes.digraph.DiGraph at 0x16506d6d0>,
<networkx.classes.digraph.DiGraph at 0x160eabe10>,
<networkx.classes.digraph.DiGraph at 0x160c372d0>,
<networkx.classes.digraph.DiGraph at 0x1658c7d10>,
<networkx.classes.digraph.DiGraph at 0x15db50e50>],
'assortativity_means': {'In-In Assortativity': -0.056004453787081056,
'In-Out Assortativity': -0.05556156322334523,
'Out-In Assortativity': -0.057581643914306986,

```

```

'Out-Out Assortativity': -0.05965027731402231},
'assortativity_stds': {'In-In Assortativity': 0.01009468985361197,
'In-Out Assortativity': 0.008416993596878954,
'Out-In Assortativity': 0.008934767507870705,
'Out-Out Assortativity': 0.007238134245412357},
'assortativity_data': {'In-In Assortativity': -0.06019317176336694,
'In-Out Assortativity': -0.060795564663496796,
'Out-In Assortativity': -0.06303814946654555,
'Out-Out Assortativity': -0.06417872159686847}},
'WDN_1996': {'nentropy_graph': <NEMtropy.graph_classes.DirectedGraph at
0x143388350>,
'fit': None,
'error_degree': 6.783182016079081e-09,
'error_strength': 369877.1213378906,
'samples': [<networkx.classes.digraph.DiGraph at 0x161eed450>,
<networkx.classes.digraph.DiGraph at 0x160842650>,
<networkx.classes.digraph.DiGraph at 0x166482410>,
<networkx.classes.digraph.DiGraph at 0x1662e9e10>,
<networkx.classes.digraph.DiGraph at 0x166ae3d50>,
<networkx.classes.digraph.DiGraph at 0x148e0a710>,
<networkx.classes.digraph.DiGraph at 0x166262bd0>,
<networkx.classes.digraph.DiGraph at 0x1667617d0>,
<networkx.classes.digraph.DiGraph at 0x166960c90>,
<networkx.classes.digraph.DiGraph at 0x166615e10>,
<networkx.classes.digraph.DiGraph at 0x161f44fd0>,
<networkx.classes.digraph.DiGraph at 0x1669637d0>,
<networkx.classes.digraph.DiGraph at 0x167c1f9d0>,
<networkx.classes.digraph.DiGraph at 0x1681238d0>,
<networkx.classes.digraph.DiGraph at 0x1683fd010>,
<networkx.classes.digraph.DiGraph at 0x16895c690>,
<networkx.classes.digraph.DiGraph at 0x1689bc9d0>,
<networkx.classes.digraph.DiGraph at 0x166761f90>,
<networkx.classes.digraph.DiGraph at 0x166628b90>,
<networkx.classes.digraph.DiGraph at 0x167985610>,
<networkx.classes.digraph.DiGraph at 0x169140690>,
<networkx.classes.digraph.DiGraph at 0x169481010>,
<networkx.classes.digraph.DiGraph at 0x169730750>,
<networkx.classes.digraph.DiGraph at 0x166262190>,
<networkx.classes.digraph.DiGraph at 0x169e157d0>,
<networkx.classes.digraph.DiGraph at 0x169efd790>,
<networkx.classes.digraph.DiGraph at 0x16a34f5d0>,
<networkx.classes.digraph.DiGraph at 0x16a982410>,
<networkx.classes.digraph.DiGraph at 0x16a6d5f10>,
<networkx.classes.digraph.DiGraph at 0x16a924b50>],
'assortativity_means': {'In-In Assortativity': -0.07116830181992069,
'In-Out Assortativity': -0.07020305753556746,
'Out-In Assortativity': -0.07007457483781604,

```

```

'Out-Out Assortativity': -0.0710156214593385},
'assortativity_stds': {'In-In Assortativity': 0.010625956705737388,
'In-Out Assortativity': 0.009627914621344246,
'Out-In Assortativity': 0.00967510543153754,
'Out-Out Assortativity': 0.008720858431337594},
'assortativity_data': {'In-In Assortativity': -0.07686964594098099,
'In-Out Assortativity': -0.07546909805023332,
'Out-In Assortativity': -0.07535502512416227,
'Out-Out Assortativity': -0.07420828746119328}},
'WDN_1995': {'nentropy_graph': <NEMtropy.graph_classes.DirectedGraph at
0x13966c410>,
'fit': None,
'error_degree': 2.3889619171768572e-09,
'error_strength': 240194.33235913515,
'samples': [<networkx.classes.digraph.DiGraph at 0x1677e4cd0>,
<networkx.classes.digraph.DiGraph at 0x16b786390>,
<networkx.classes.digraph.DiGraph at 0x1667553d0>,
<networkx.classes.digraph.DiGraph at 0x16b1e0e50>,
<networkx.classes.digraph.DiGraph at 0x16b6ab350>,
<networkx.classes.digraph.DiGraph at 0x16ba471d0>,
<networkx.classes.digraph.DiGraph at 0x16b595f10>,
<networkx.classes.digraph.DiGraph at 0x16b16f2d0>,
<networkx.classes.digraph.DiGraph at 0x16c25c5d0>,
<networkx.classes.digraph.DiGraph at 0x16b7f9990>,
<networkx.classes.digraph.DiGraph at 0x16c4f5410>,
<networkx.classes.digraph.DiGraph at 0x16b30b2d0>,
<networkx.classes.digraph.DiGraph at 0x16b16bf10>,
<networkx.classes.digraph.DiGraph at 0x16cd7cf10>,
<networkx.classes.digraph.DiGraph at 0x16b75be10>,
<networkx.classes.digraph.DiGraph at 0x16d51c750>,
<networkx.classes.digraph.DiGraph at 0x16c718790>,
<networkx.classes.digraph.DiGraph at 0x16b328e90>,
<networkx.classes.digraph.DiGraph at 0x16dc3c8d0>,
<networkx.classes.digraph.DiGraph at 0x16dc3cf90>,
<networkx.classes.digraph.DiGraph at 0x16b30a250>,
<networkx.classes.digraph.DiGraph at 0x15f418490>,
<networkx.classes.digraph.DiGraph at 0x16b38e310>,
<networkx.classes.digraph.DiGraph at 0x16e918750>,
<networkx.classes.digraph.DiGraph at 0x16ea9fd90>,
<networkx.classes.digraph.DiGraph at 0x16ea9ff10>,
<networkx.classes.digraph.DiGraph at 0x16eb7ee50>,
<networkx.classes.digraph.DiGraph at 0x14c7805d0>,
<networkx.classes.digraph.DiGraph at 0x16b46a310>,
<networkx.classes.digraph.DiGraph at 0x16f277dd0>],
'assortativity_means': {'In-In Assortativity': -0.07834395261485842,
'In-Out Assortativity': -0.07608028990962798,
'Out-In Assortativity': -0.07758440580884819,

```

```

'Out-Out Assortativity': -0.07710979425832175},
'assortativity_stds': {'In-In Assortativity': 0.0119364311205388,
'In-Out Assortativity': 0.01235769486498981,
'Out-In Assortativity': 0.012742341037601067,
'Out-Out Assortativity': 0.013597466061968062},
'assortativity_data': {'In-In Assortativity': -0.07921393553292572,
'In-Out Assortativity': -0.07702100817504068,
'Out-In Assortativity': -0.07830645801950578,
'Out-Out Assortativity': -0.07638722326404519}},
'WDN_1999': {'nentropy_graph': <NEMtropy.graph_classes.DirectedGraph at
0x13966be10>,
'fit': None,
'error_degree': 1.485290113123483e-09,
'error_strength': 181065.18603515625,
'samples': [<networkx.classes.digraph.DiGraph at 0x16b27a610>,
<networkx.classes.digraph.DiGraph at 0x16ff83c90>,
<networkx.classes.digraph.DiGraph at 0x170190a50>,
<networkx.classes.digraph.DiGraph at 0x1702250d0>,
<networkx.classes.digraph.DiGraph at 0x16ff92550>,
<networkx.classes.digraph.DiGraph at 0x170307f50>,
<networkx.classes.digraph.DiGraph at 0x16fd472d0>,
<networkx.classes.digraph.DiGraph at 0x170cb09d0>,
<networkx.classes.digraph.DiGraph at 0x170e6b410>,
<networkx.classes.digraph.DiGraph at 0x16f7ee810>,
<networkx.classes.digraph.DiGraph at 0x171870710>,
<networkx.classes.digraph.DiGraph at 0x16ff9bd50>,
<networkx.classes.digraph.DiGraph at 0x171b3c890>,
<networkx.classes.digraph.DiGraph at 0x170cc0790>,
<networkx.classes.digraph.DiGraph at 0x1724a0690>,
<networkx.classes.digraph.DiGraph at 0x17246c5d0>,
<networkx.classes.digraph.DiGraph at 0x172444950>,
<networkx.classes.digraph.DiGraph at 0x171828a10>,
<networkx.classes.digraph.DiGraph at 0x1717fadd0>,
<networkx.classes.digraph.DiGraph at 0x171b14910>,
<networkx.classes.digraph.DiGraph at 0x1733105d0>,
<networkx.classes.digraph.DiGraph at 0x1739248d0>,
<networkx.classes.digraph.DiGraph at 0x173d90690>,
<networkx.classes.digraph.DiGraph at 0x173c68a90>,
<networkx.classes.digraph.DiGraph at 0x14b9e4e10>,
<networkx.classes.digraph.DiGraph at 0x1740dde50>,
<networkx.classes.digraph.DiGraph at 0x174400590>,
<networkx.classes.digraph.DiGraph at 0x174674410>,
<networkx.classes.digraph.DiGraph at 0x146490dd0>,
<networkx.classes.digraph.DiGraph at 0x17498c4d0>],
'assortativity_means': {'In-In Assortativity': -0.06258685262470548,
'In-Out Assortativity': -0.062026375526850794,
'Out-In Assortativity': -0.062467242717497824,

```

```

'Out-Out Assortativity': -0.06386361819527316},
'assortativity_stds': {'In-In Assortativity': 0.011229386509438447,
'In-Out Assortativity': 0.010785805555080321,
'Out-In Assortativity': 0.010928589487020975,
'Out-Out Assortativity': 0.010696098142217888},
'assortativity_data': {'In-In Assortativity': -0.06360116568177393,
'In-Out Assortativity': -0.06507198196482604,
'Out-In Assortativity': -0.06546711179930208,
'Out-Out Assortativity': -0.06730720421208614}},
'WDN_2001': {'nentropy_graph': <NEMtropy.graph_classes.DirectedGraph at
0x143770110>,
'fit': None,
'error_degree': 3.313960661444071e-09,
'error_strength': 435455.66259765625,
'samples': [<networkx.classes.digraph.DiGraph at 0x16fdc4150>,
<networkx.classes.digraph.DiGraph at 0x175c7b5d0>,
<networkx.classes.digraph.DiGraph at 0x17526ad10>,
<networkx.classes.digraph.DiGraph at 0x16fd45a10>,
<networkx.classes.digraph.DiGraph at 0x175f98cd0>,
<networkx.classes.digraph.DiGraph at 0x175f98510>,
<networkx.classes.digraph.DiGraph at 0x175c7bfd0>,
<networkx.classes.digraph.DiGraph at 0x176580a90>,
<networkx.classes.digraph.DiGraph at 0x176896850>,
<networkx.classes.digraph.DiGraph at 0x175c93d50>,
<networkx.classes.digraph.DiGraph at 0x1772cead0>,
<networkx.classes.digraph.DiGraph at 0x1772e08d0>,
<networkx.classes.digraph.DiGraph at 0x1756ce450>,
<networkx.classes.digraph.DiGraph at 0x177e38810>,
<networkx.classes.digraph.DiGraph at 0x1782d06d0>,
<networkx.classes.digraph.DiGraph at 0x1782bca90>,
<networkx.classes.digraph.DiGraph at 0x1782e0390>,
<networkx.classes.digraph.DiGraph at 0x1788e6d10>,
<networkx.classes.digraph.DiGraph at 0x178adf6d0>,
<networkx.classes.digraph.DiGraph at 0x16ed31fd0>,
<networkx.classes.digraph.DiGraph at 0x1787d7590>,
<networkx.classes.digraph.DiGraph at 0x175c93c90>,
<networkx.classes.digraph.DiGraph at 0x1798949d0>,
<networkx.classes.digraph.DiGraph at 0x1798e8390>,
<networkx.classes.digraph.DiGraph at 0x17a164890>,
<networkx.classes.digraph.DiGraph at 0x179ebc650>,
<networkx.classes.digraph.DiGraph at 0x175d09350>,
<networkx.classes.digraph.DiGraph at 0x175c194d0>,
<networkx.classes.digraph.DiGraph at 0x175c78290>,
<networkx.classes.digraph.DiGraph at 0x15ecfba10>],
'assortativity_means': {'In-In Assortativity': -0.05797759295899372,
'In-Out Assortativity': -0.05777682291226383,
'Out-In Assortativity': -0.058476908288988855,

```

```

'Out-Out Assortativity': -0.060480652926127434},
'assortativity_stds': {'In-In Assortativity': 0.009368840037343541,
'In-Out Assortativity': 0.009531801002439723,
'Out-In Assortativity': 0.009555005258801434,
'Out-Out Assortativity': 0.010341923521694489},
'assortativity_data': {'In-In Assortativity': -0.060016931181439886,
'In-Out Assortativity': -0.061178239732267325,
'Out-In Assortativity': -0.06192625961852602,
'Out-Out Assortativity': -0.06348734473716487}}}}

```

```

[185]: for graph_name, results in nemtropy_undirected_results.items():

    graph = results['nemtropy_graph']
    error_degree = results['error_degree']
    error_strength = results['error_strength']

    # Generate 30 ensembles copies using ensemble_sampler
    graph.ensemble_sampler(SAMPLE_SIZE, cpu_n=2,
↪output_dir=f"{OUTPUT_PATH_ERGMS}/{graph_name}-undirected-sample/")

    sampled_graphs = []
    assortativity_values = []
    for i in range(0, SAMPLE_SIZE, 1):
        # We read the edgelist
        edgelist_ens = np.loadtxt(f"{OUTPUT_PATH_ERGMS}/{
↪graph_name}-undirected-sample/{i}.txt")

        # and build the adjacency matrix
        ens_adj = build_adjacency_from_edgelist(edgelist = edgelist_ens,
                                                is_directed = True,
                                                is_sparse = False,
                                                is_weighted = True)

        sampled_graph = to_networkx_graph(ens_adj)
        sampled_graphs.append(sampled_graph)

        # measure strength assortativity
        assortativities_sampled = calculate_assortativity(sampled_graph)
        assortativity_values.append(assortativities_sampled)

    categories = list(assortativity_values[0].keys())

    # Initialize dictionaries to store mean and standard deviation for each
↪category
    mean_values = {}
    std_values = {}

```

```

# Loop through each category
for category in categories:
    # Extract values for the current category across all dictionaries
    category_values = [data[category] for data in assortativity_values]

    # Calculate mean and standard
    mean = np.mean(category_values)
    std = np.std(category_values)

    # Store mean and standard deviation for the category
    mean_values[category] = mean
    std_values[category] = std

results['samples'] = sampled_graphs
results['assortativity_means'] = mean_values
results['assortativity_stds'] = std_values
results['assortativity_data'] = assortativities[graph_name]

```

```
[186]: nemtropy_undirected_results
```

```

[186]: {'WDN_1997': {'nemtropy_graph': <NEMtropy.graph_classes.UndirectedGraph at
0x1461904d0>,
'fit': None,
'error_degree': 2.2947688194108196e-09,
'error_strength': None,
'samples': [<networkx.classes.digraph.DiGraph at 0x176bfde90>,
<networkx.classes.digraph.DiGraph at 0x17b7200d0>,
<networkx.classes.digraph.DiGraph at 0x17b767850>,
<networkx.classes.digraph.DiGraph at 0x17b670650>,
<networkx.classes.digraph.DiGraph at 0x17b697a90>,
<networkx.classes.digraph.DiGraph at 0x1490d7490>,
<networkx.classes.digraph.DiGraph at 0x17be83950>,
<networkx.classes.digraph.DiGraph at 0x17b89d8d0>,
<networkx.classes.digraph.DiGraph at 0x14f303850>,
<networkx.classes.digraph.DiGraph at 0x17bb03a10>,
<networkx.classes.digraph.DiGraph at 0x17c5acbd0>,
<networkx.classes.digraph.DiGraph at 0x17c5809d0>,
<networkx.classes.digraph.DiGraph at 0x17c768690>,
<networkx.classes.digraph.DiGraph at 0x17c58bb10>,
<networkx.classes.digraph.DiGraph at 0x17ccbfed0>,
<networkx.classes.digraph.DiGraph at 0x17ceb7f50>,
<networkx.classes.digraph.DiGraph at 0x17ceb7050>,
<networkx.classes.digraph.DiGraph at 0x17d4dcd10>,
<networkx.classes.digraph.DiGraph at 0x17b695a50>,
<networkx.classes.digraph.DiGraph at 0x17d46cf90>,
<networkx.classes.digraph.DiGraph at 0x17d6d3890>,
<networkx.classes.digraph.DiGraph at 0x17d85d410>,

```

```

<networkx.classes.digraph.DiGraph at 0x17dbb4950>,
<networkx.classes.digraph.DiGraph at 0x17df6bed0>,
<networkx.classes.digraph.DiGraph at 0x17bca3010>,
<networkx.classes.digraph.DiGraph at 0x17df90950>,
<networkx.classes.digraph.DiGraph at 0x17e326790>,
<networkx.classes.digraph.DiGraph at 0x17b89d590>,
<networkx.classes.digraph.DiGraph at 0x17e509790>,
<networkx.classes.digraph.DiGraph at 0x17e8bbc10>],
'assortativity_means': {'In-In Assortativity': 0.3356737136368659,
'In-Out Assortativity': -0.315950525819607,
'Out-In Assortativity': 0.09933031713893721,
'Out-Out Assortativity': -0.0981412305255458},
'assortativity_std': {'In-In Assortativity': 0.06758472472796187,
'In-Out Assortativity': 0.10921065164944739,
'Out-In Assortativity': 0.0078652281610141,
'Out-Out Assortativity': 0.026267451135385065},
'assortativity_data': {'In-In Assortativity': -0.07368635207387701,
'In-Out Assortativity': -0.07132955497657792,
'Out-In Assortativity': -0.07171683897072376,
'Out-Out Assortativity': -0.06970079872009399}},
'WDN_1992': {'nentropy_graph': <NEMtropy.graph_classes.UndirectedGraph at
0x1313f6c10>,
'fit': None,
'error_degree': 6.044842848496046e-09,
'error_strength': None,
'samples': [<networkx.classes.digraph.DiGraph at 0x17b529a90>,
<networkx.classes.digraph.DiGraph at 0x17ef08190>,
<networkx.classes.digraph.DiGraph at 0x17ef33150>,
<networkx.classes.digraph.DiGraph at 0x17eed7190>,
<networkx.classes.digraph.DiGraph at 0x17ed558d0>,
<networkx.classes.digraph.DiGraph at 0x17edc1210>,
<networkx.classes.digraph.DiGraph at 0x17ee1dc10>,
<networkx.classes.digraph.DiGraph at 0x17f484e10>,
<networkx.classes.digraph.DiGraph at 0x17f36ff90>,
<networkx.classes.digraph.DiGraph at 0x17f487950>,
<networkx.classes.digraph.DiGraph at 0x17eed5450>,
<networkx.classes.digraph.DiGraph at 0x17f59c210>,
<networkx.classes.digraph.DiGraph at 0x17f8c8a90>,
<networkx.classes.digraph.DiGraph at 0x17faece10>,
<networkx.classes.digraph.DiGraph at 0x17e5e4f10>,
<networkx.classes.digraph.DiGraph at 0x17ef04290>,
<networkx.classes.digraph.DiGraph at 0x17eed7790>,
<networkx.classes.digraph.DiGraph at 0x17ff48e10>,
<networkx.classes.digraph.DiGraph at 0x17ee1c9d0>,
<networkx.classes.digraph.DiGraph at 0x17f01e250>,
<networkx.classes.digraph.DiGraph at 0x17edbd010>,
<networkx.classes.digraph.DiGraph at 0x18005f850>,

```

```

<networkx.classes.digraph.DiGraph at 0x17ee1ffd0>,
<networkx.classes.digraph.DiGraph at 0x17f254890>,
<networkx.classes.digraph.DiGraph at 0x17b74ddd0>,
<networkx.classes.digraph.DiGraph at 0x17e988c10>,
<networkx.classes.digraph.DiGraph at 0x17eed6090>,
<networkx.classes.digraph.DiGraph at 0x1809f60d0>,
<networkx.classes.digraph.DiGraph at 0x180b3cc90>,
<networkx.classes.digraph.DiGraph at 0x180c58d90>],
'assortativity_means': {'In-In Assortativity': 0.08963012151336258,
'In-Out Assortativity': -0.34682576689264294,
'Out-In Assortativity': -0.17463592764922214,
'Out-Out Assortativity': -0.3137006302391205},
'assortativity_std': {'In-In Assortativity': 0.004599489581603946,
'In-Out Assortativity': 0.0034496125542308183,
'Out-In Assortativity': 0.004915827860251767,
'Out-Out Assortativity': 0.0048326300893460385},
'assortativity_data': {'In-In Assortativity': -0.04868776409103533,
'In-Out Assortativity': -0.046647404340923876,
'Out-In Assortativity': -0.059869793086991335,
'Out-Out Assortativity': -0.05760318920541264}},
'WDN_2000': {'nentropy_graph': <NEMtropy.graph_classes.UndirectedGraph at
0x1461919d0>,
'fit': None,
'error_degree': 2.2773178898205515e-09,
'error_strength': None,
'samples': [<networkx.classes.digraph.DiGraph at 0x17f202f90>,
<networkx.classes.digraph.DiGraph at 0x17f6bc1d0>,
<networkx.classes.digraph.DiGraph at 0x180d2c090>,
<networkx.classes.digraph.DiGraph at 0x180e93950>,
<networkx.classes.digraph.DiGraph at 0x18132db50>,
<networkx.classes.digraph.DiGraph at 0x1815296d0>,
<networkx.classes.digraph.DiGraph at 0x180dfbdd0>,
<networkx.classes.digraph.DiGraph at 0x18177c8d0>,
<networkx.classes.digraph.DiGraph at 0x1808f45d0>,
<networkx.classes.digraph.DiGraph at 0x181cfdc10>,
<networkx.classes.digraph.DiGraph at 0x181175050>,
<networkx.classes.digraph.DiGraph at 0x1820f5350>,
<networkx.classes.digraph.DiGraph at 0x181f2ab90>,
<networkx.classes.digraph.DiGraph at 0x17edff610>,
<networkx.classes.digraph.DiGraph at 0x1827a6690>,
<networkx.classes.digraph.DiGraph at 0x182716d10>,
<networkx.classes.digraph.DiGraph at 0x182adc910>,
<networkx.classes.digraph.DiGraph at 0x1827721d0>,
<networkx.classes.digraph.DiGraph at 0x182b3bcd0>,
<networkx.classes.digraph.DiGraph at 0x18101d450>,
<networkx.classes.digraph.DiGraph at 0x154ab94d0>,
<networkx.classes.digraph.DiGraph at 0x183541b10>,

```

```

<networkx.classes.digraph.DiGraph at 0x18379d6d0>,
<networkx.classes.digraph.DiGraph at 0x183994190>,
<networkx.classes.digraph.DiGraph at 0x183accb10>,
<networkx.classes.digraph.DiGraph at 0x183cb8c50>,
<networkx.classes.digraph.DiGraph at 0x183eabfd0>,
<networkx.classes.digraph.DiGraph at 0x183f79210>,
<networkx.classes.digraph.DiGraph at 0x1840cfd50>,
<networkx.classes.digraph.DiGraph at 0x181075dd0>],
'assortativity_means': {'In-In Assortativity': 0.349693152210142,
'In-Out Assortativity': -0.30181505130904257,
'Out-In Assortativity': 0.13600406588330527,
'Out-Out Assortativity': -0.07095587124366008},
'assortativity_std': {'In-In Assortativity': 0.08469101058745171,
'In-Out Assortativity': 0.12983899517698672,
'Out-In Assortativity': 0.011454566207095689,
'Out-Out Assortativity': 0.023377836708618833},
'assortativity_data': {'In-In Assortativity': -0.0631970717497539,
'In-Out Assortativity': -0.06467170213846346,
'Out-In Assortativity': -0.06539073425903164,
'Out-Out Assortativity': -0.06730072607061888}},
'WDN_1998': {'nentropy_graph': <NEMtropy.graph_classes.UndirectedGraph at
0x1439a6e50>,
'fit': None,
'error_degree': 1.6948717984632822e-09,
'error_strength': None,
'samples': [<networkx.classes.digraph.DiGraph at 0x181170750>,
<networkx.classes.digraph.DiGraph at 0x184ca02d0>,
<networkx.classes.digraph.DiGraph at 0x184980950>,
<networkx.classes.digraph.DiGraph at 0x184ce9210>,
<networkx.classes.digraph.DiGraph at 0x1849490d0>,
<networkx.classes.digraph.DiGraph at 0x18511dc50>,
<networkx.classes.digraph.DiGraph at 0x1852f3110>,
<networkx.classes.digraph.DiGraph at 0x185278610>,
<networkx.classes.digraph.DiGraph at 0x1856b0c50>,
<networkx.classes.digraph.DiGraph at 0x1857f4a90>,
<networkx.classes.digraph.DiGraph at 0x185a24150>,
<networkx.classes.digraph.DiGraph at 0x185a55410>,
<networkx.classes.digraph.DiGraph at 0x185d9dc10>,
<networkx.classes.digraph.DiGraph at 0x185db5e50>,
<networkx.classes.digraph.DiGraph at 0x186172010>,
<networkx.classes.digraph.DiGraph at 0x186365010>,
<networkx.classes.digraph.DiGraph at 0x18489f290>,
<networkx.classes.digraph.DiGraph at 0x185828e50>,
<networkx.classes.digraph.DiGraph at 0x1857e1ad0>,
<networkx.classes.digraph.DiGraph at 0x18132e190>,
<networkx.classes.digraph.DiGraph at 0x186dc3d90>,
<networkx.classes.digraph.DiGraph at 0x186ed0550>,

```

```

<networkx.classes.digraph.DiGraph at 0x1870b7b10>,
<networkx.classes.digraph.DiGraph at 0x1870b73d0>,
<networkx.classes.digraph.DiGraph at 0x187369310>,
<networkx.classes.digraph.DiGraph at 0x187514f90>,
<networkx.classes.digraph.DiGraph at 0x184c06090>,
<networkx.classes.digraph.DiGraph at 0x1869c36d0>,
<networkx.classes.digraph.DiGraph at 0x187c31110>,
<networkx.classes.digraph.DiGraph at 0x17f203310>],
'assortativity_means': {'In-In Assortativity': 0.3677309298793979,
'In-Out Assortativity': -0.34462289619075404,
'Out-In Assortativity': 0.11566476419242777,
'Out-Out Assortativity': -0.09126562190943617},
'assortativity_std': {'In-In Assortativity': 0.002976815376083894,
'In-Out Assortativity': 0.003459164896120447,
'Out-In Assortativity': 0.005455110227211449,
'Out-Out Assortativity': 0.004208806561630391},
'assortativity_data': {'In-In Assortativity': -0.0655705781839572,
'In-Out Assortativity': -0.06540616504184192,
'Out-In Assortativity': -0.06572057397949527,
'Out-Out Assortativity': -0.06583617855060356}},
'WDN_1994': {'nentropy_graph': <NEMtropy.graph_classes.UndirectedGraph at
0x1434ac990>,
'fit': None,
'error_degree': 2.8035458399244817e-09,
'error_strength': None,
'samples': [<networkx.classes.digraph.DiGraph at 0x185247ed0>,
<networkx.classes.digraph.DiGraph at 0x184ed47d0>,
<networkx.classes.digraph.DiGraph at 0x184ca2d50>,
<networkx.classes.digraph.DiGraph at 0x1884fecdd0>,
<networkx.classes.digraph.DiGraph at 0x1883a8ad0>,
<networkx.classes.digraph.DiGraph at 0x1883b3b90>,
<networkx.classes.digraph.DiGraph at 0x18837acd0>,
<networkx.classes.digraph.DiGraph at 0x1889c5510>,
<networkx.classes.digraph.DiGraph at 0x188b12250>,
<networkx.classes.digraph.DiGraph at 0x188e5bf50>,
<networkx.classes.digraph.DiGraph at 0x188c9d4d0>,
<networkx.classes.digraph.DiGraph at 0x17f0cbdd0>,
<networkx.classes.digraph.DiGraph at 0x1890105d0>,
<networkx.classes.digraph.DiGraph at 0x189154410>,
<networkx.classes.digraph.DiGraph at 0x1894aa450>,
<networkx.classes.digraph.DiGraph at 0x1898f4950>,
<networkx.classes.digraph.DiGraph at 0x1898f4f90>,
<networkx.classes.digraph.DiGraph at 0x188289fd0>,
<networkx.classes.digraph.DiGraph at 0x189c03810>,
<networkx.classes.digraph.DiGraph at 0x1850d34d0>,
<networkx.classes.digraph.DiGraph at 0x189f29a90>,
<networkx.classes.digraph.DiGraph at 0x18a072410>,

```

```

<networkx.classes.digraph.DiGraph at 0x18a0b9150>,
<networkx.classes.digraph.DiGraph at 0x18a25d810>,
<networkx.classes.digraph.DiGraph at 0x18a50aed0>,
<networkx.classes.digraph.DiGraph at 0x188289550>,
<networkx.classes.digraph.DiGraph at 0x18a6a49d0>,
<networkx.classes.digraph.DiGraph at 0x18a99ae90>,
<networkx.classes.digraph.DiGraph at 0x188379350>,
<networkx.classes.digraph.DiGraph at 0x18aca7750>],
'assortativity_means': {'In-In Assortativity': 0.24888486085665779,
'In-Out Assortativity': -0.35048750587458416,
'Out-In Assortativity': -0.02229325634958699,
'Out-Out Assortativity': -0.2007817750529001},
'assortativity_stds': {'In-In Assortativity': 0.003947596258236475,
'In-Out Assortativity': 0.003692448207533778,
'Out-In Assortativity': 0.005141406454122883,
'Out-Out Assortativity': 0.005149888140169398},
'assortativity_data': {'In-In Assortativity': -0.07868988970611883,
'In-Out Assortativity': -0.07622523439269807,
'Out-In Assortativity': -0.08022501617948047,
'Out-Out Assortativity': -0.07800143058821393}},
'WDN_1993': {'nentropy_graph': <NEMtropy.graph_classes.UndirectedGraph at
0x13189f410>,
'fit': None,
'error_degree': 7.512738875448122e-09,
'error_strength': None,
'samples': [<networkx.classes.digraph.DiGraph at 0x18ae73ad0>,
<networkx.classes.digraph.DiGraph at 0x1881e7710>,
<networkx.classes.digraph.DiGraph at 0x1881e7c90>,
<networkx.classes.digraph.DiGraph at 0x18b42cd50>,
<networkx.classes.digraph.DiGraph at 0x18b0a3150>,
<networkx.classes.digraph.DiGraph at 0x18b03e910>,
<networkx.classes.digraph.DiGraph at 0x18b69f2d0>,
<networkx.classes.digraph.DiGraph at 0x18b7b7d10>,
<networkx.classes.digraph.DiGraph at 0x18b69b290>,
<networkx.classes.digraph.DiGraph at 0x18b13b290>,
<networkx.classes.digraph.DiGraph at 0x18b2c1a90>,
<networkx.classes.digraph.DiGraph at 0x18bcdd790>,
<networkx.classes.digraph.DiGraph at 0x18be2e8d0>,
<networkx.classes.digraph.DiGraph at 0x18bea52d0>,
<networkx.classes.digraph.DiGraph at 0x18c0bac50>,
<networkx.classes.digraph.DiGraph at 0x18c224990>,
<networkx.classes.digraph.DiGraph at 0x18c22d110>,
<networkx.classes.digraph.DiGraph at 0x18c3d6390>,
<networkx.classes.digraph.DiGraph at 0x18b173990>,
<networkx.classes.digraph.DiGraph at 0x18c745990>,
<networkx.classes.digraph.DiGraph at 0x18b0a2c90>,
<networkx.classes.digraph.DiGraph at 0x18b062750>,

```

```

<networkx.classes.digraph.DiGraph at 0x18b0625d0>,
<networkx.classes.digraph.DiGraph at 0x18cb2eb90>,
<networkx.classes.digraph.DiGraph at 0x18ccd5490>,
<networkx.classes.digraph.DiGraph at 0x18cefcc50>,
<networkx.classes.digraph.DiGraph at 0x18cdb9710>,
<networkx.classes.digraph.DiGraph at 0x18d194fd0>,
<networkx.classes.digraph.DiGraph at 0x18d0b7b50>,
<networkx.classes.digraph.DiGraph at 0x18d431550>],
'assortativity_means': {'In-In Assortativity': 0.14735550415684015,
'In-Out Assortativity': -0.34329370591795794,
'Out-In Assortativity': -0.12127136015117948,
'Out-Out Assortativity': -0.2742455288869968},
'assortativity_std': {'In-In Assortativity': 0.004064769433040502,
'In-Out Assortativity': 0.003659331223726577,
'Out-In Assortativity': 0.004355693146764099,
'Out-Out Assortativity': 0.0050014861777792265},
'assortativity_data': {'In-In Assortativity': -0.06537929984057925,
'In-Out Assortativity': -0.062189867187614475,
'Out-In Assortativity': -0.07202068872957715,
'Out-Out Assortativity': -0.06910618411247677}},
'WDN_2002': {'nentropy_graph': <NEMtropy.graph_classes.UndirectedGraph at
0x143838150>,
'fit': None,
'error_degree': 2.058698100881884e-09,
'error_strength': None,
'samples': [<networkx.classes.digraph.DiGraph at 0x18b65f390>,
<networkx.classes.digraph.DiGraph at 0x18bb8ef90>,
<networkx.classes.digraph.DiGraph at 0x18114cd50>,
<networkx.classes.digraph.DiGraph at 0x18dcd5a50>,
<networkx.classes.digraph.DiGraph at 0x142e3bbd0>,
<networkx.classes.digraph.DiGraph at 0x18deeb0d0>,
<networkx.classes.digraph.DiGraph at 0x18e139ad0>,
<networkx.classes.digraph.DiGraph at 0x18e30bfd0>,
<networkx.classes.digraph.DiGraph at 0x18d8eaed0>,
<networkx.classes.digraph.DiGraph at 0x18e704590>,
<networkx.classes.digraph.DiGraph at 0x18e96b790>,
<networkx.classes.digraph.DiGraph at 0x18eb67e50>,
<networkx.classes.digraph.DiGraph at 0x18ead5b90>,
<networkx.classes.digraph.DiGraph at 0x18db2c450>,
<networkx.classes.digraph.DiGraph at 0x18f139e90>,
<networkx.classes.digraph.DiGraph at 0x18f2a3990>,
<networkx.classes.digraph.DiGraph at 0x18f4f8650>,
<networkx.classes.digraph.DiGraph at 0x18f69fa10>,
<networkx.classes.digraph.DiGraph at 0x18f6b6a50>,
<networkx.classes.digraph.DiGraph at 0x18f8e0610>,
<networkx.classes.digraph.DiGraph at 0x18fc95310>,
<networkx.classes.digraph.DiGraph at 0x18f0ffad0>,

```

```

<networkx.classes.digraph.DiGraph at 0x18ef73fd0>,
<networkx.classes.digraph.DiGraph at 0x18f0b7410>,
<networkx.classes.digraph.DiGraph at 0x1902c8110>,
<networkx.classes.digraph.DiGraph at 0x19067b110>,
<networkx.classes.digraph.DiGraph at 0x19087b090>,
<networkx.classes.digraph.DiGraph at 0x190a76d50>,
<networkx.classes.digraph.DiGraph at 0x190c763d0>,
<networkx.classes.digraph.DiGraph at 0x190e6e6d0>],
'assortativity_means': {'In-In Assortativity': 0.3827008810388511,
'In-Out Assortativity': -0.3482982731037963,
'Out-In Assortativity': 0.13330474453521343,
'Out-Out Assortativity': -0.0868423177789473},
'assortativity_stds': {'In-In Assortativity': 0.0031450866260855654,
'In-Out Assortativity': 0.002921424880223263,
'Out-In Assortativity': 0.00504076026492231,
'Out-Out Assortativity': 0.004622001303048389},
'assortativity_data': {'In-In Assortativity': -0.06019317176336694,
'In-Out Assortativity': -0.060795564663496796,
'Out-In Assortativity': -0.06303814946654555,
'Out-Out Assortativity': -0.06417872159686847}},
'WDN_1996': {'nentropy_graph': <NEMtropy.graph_classes.UndirectedGraph at
0x1395ac810>,
'fit': None,
'error_degree': 1.957971562660532e-09,
'error_strength': None,
'samples': [<networkx.classes.digraph.DiGraph at 0x18b2a5590>,
<networkx.classes.digraph.DiGraph at 0x18e335ed0>,
<networkx.classes.digraph.DiGraph at 0x191504050>,
<networkx.classes.digraph.DiGraph at 0x1914141d0>,
<networkx.classes.digraph.DiGraph at 0x1883b3210>,
<networkx.classes.digraph.DiGraph at 0x1919a1fd0>,
<networkx.classes.digraph.DiGraph at 0x191582e50>,
<networkx.classes.digraph.DiGraph at 0x191d2cc90>,
<networkx.classes.digraph.DiGraph at 0x191eed7d0>,
<networkx.classes.digraph.DiGraph at 0x1914bea10>,
<networkx.classes.digraph.DiGraph at 0x1924b5050>,
<networkx.classes.digraph.DiGraph at 0x1924b5790>,
<networkx.classes.digraph.DiGraph at 0x1925df8d0>,
<networkx.classes.digraph.DiGraph at 0x1927a7750>,
<networkx.classes.digraph.DiGraph at 0x192b38e50>,
<networkx.classes.digraph.DiGraph at 0x192b39210>,
<networkx.classes.digraph.DiGraph at 0x192b48850>,
<networkx.classes.digraph.DiGraph at 0x192f8af10>,
<networkx.classes.digraph.DiGraph at 0x192daaa10>,
<networkx.classes.digraph.DiGraph at 0x1930b3e10>,
<networkx.classes.digraph.DiGraph at 0x193417bd0>,
<networkx.classes.digraph.DiGraph at 0x1935d7d10>,

```

```

<networkx.classes.digraph.DiGraph at 0x191583e10>,
<networkx.classes.digraph.DiGraph at 0x193984b50>,
<networkx.classes.digraph.DiGraph at 0x1464d2a10>,
<networkx.classes.digraph.DiGraph at 0x193bebe10>,
<networkx.classes.digraph.DiGraph at 0x1916d9d10>,
<networkx.classes.digraph.DiGraph at 0x193ed49d0>,
<networkx.classes.digraph.DiGraph at 0x19424fd10>,
<networkx.classes.digraph.DiGraph at 0x19426fbd0>],
'assortativity_means': {'In-In Assortativity': 0.3288065003077054,
'In-Out Assortativity': -0.34419232144406625,
'Out-In Assortativity': 0.0756220464173495,
'Out-Out Assortativity': -0.11410647465928915},
'assortativity_std': {'In-In Assortativity': 0.003260123677209303,
'In-Out Assortativity': 0.002271362099875487,
'Out-In Assortativity': 0.005313021261929174,
'Out-Out Assortativity': 0.003942511754101271},
'assortativity_data': {'In-In Assortativity': -0.07686964594098099,
'In-Out Assortativity': -0.07546909805023332,
'Out-In Assortativity': -0.07535502512416227,
'Out-Out Assortativity': -0.07420828746119328}},
'WDN_1995': {'nentropy_graph': <NEMtropy.graph_classes.UndirectedGraph at
0x1438f8350>,
'fit': None,
'error_degree': 2.09220729630033e-09,
'error_strength': None,
'samples': [<networkx.classes.digraph.DiGraph at 0x1922f8c50>,
<networkx.classes.digraph.DiGraph at 0x194857750>,
<networkx.classes.digraph.DiGraph at 0x194b8ae10>,
<networkx.classes.digraph.DiGraph at 0x194ba9090>,
<networkx.classes.digraph.DiGraph at 0x194d02c90>,
<networkx.classes.digraph.DiGraph at 0x194d01c50>,
<networkx.classes.digraph.DiGraph at 0x1950ba1d0>,
<networkx.classes.digraph.DiGraph at 0x194e83150>,
<networkx.classes.digraph.DiGraph at 0x195420350>,
<networkx.classes.digraph.DiGraph at 0x1955539d0>,
<networkx.classes.digraph.DiGraph at 0x17d6f0b10>,
<networkx.classes.digraph.DiGraph at 0x1957317d0>,
<networkx.classes.digraph.DiGraph at 0x195a6b650>,
<networkx.classes.digraph.DiGraph at 0x195dd0a90>,
<networkx.classes.digraph.DiGraph at 0x19153dfd0>,
<networkx.classes.digraph.DiGraph at 0x195f82b90>,
<networkx.classes.digraph.DiGraph at 0x195fe1590>,
<networkx.classes.digraph.DiGraph at 0x1961b5f90>,
<networkx.classes.digraph.DiGraph at 0x194b1a310>,
<networkx.classes.digraph.DiGraph at 0x1964e9990>,
<networkx.classes.digraph.DiGraph at 0x194828bd0>,
<networkx.classes.digraph.DiGraph at 0x1969853d0>,

```

```

<networkx.classes.digraph.DiGraph at 0x1969a3ed0>,
<networkx.classes.digraph.DiGraph at 0x196ce2210>,
<networkx.classes.digraph.DiGraph at 0x196eb07d0>,
<networkx.classes.digraph.DiGraph at 0x196edf090>,
<networkx.classes.digraph.DiGraph at 0x194969810>,
<networkx.classes.digraph.DiGraph at 0x1973b1e90>,
<networkx.classes.digraph.DiGraph at 0x19483d990>,
<networkx.classes.digraph.DiGraph at 0x1948f7cd0>],
'assortativity_means': {'In-In Assortativity': 0.31412419783297724,
'In-Out Assortativity': -0.3334055122594612,
'Out-In Assortativity': 0.031094106770170597,
'Out-Out Assortativity': -0.13078437044746954},
'assortativity_std': {'In-In Assortativity': 0.0037996178050235556,
'In-Out Assortativity': 0.0031277827904254963,
'Out-In Assortativity': 0.005946135359184433,
'Out-Out Assortativity': 0.004181904779145818},
'assortativity_data': {'In-In Assortativity': -0.07921393553292572,
'In-Out Assortativity': -0.07702100817504068,
'Out-In Assortativity': -0.07830645801950578,
'Out-Out Assortativity': -0.07638722326404519}},
'WDN_1999': {'nentropy_graph': <NEMtropy.graph_classes.UndirectedGraph at
0x14346cdd0>,
'fit': None,
'error_degree': 8.603535661677597e-10,
'error_strength': None,
'samples': [<networkx.classes.digraph.DiGraph at 0x194bbc1d0>,
<networkx.classes.digraph.DiGraph at 0x195706510>,
<networkx.classes.digraph.DiGraph at 0x194baa1d0>,
<networkx.classes.digraph.DiGraph at 0x197ca8f10>,
<networkx.classes.digraph.DiGraph at 0x197ca8b50>,
<networkx.classes.digraph.DiGraph at 0x198263d10>,
<networkx.classes.digraph.DiGraph at 0x197ce8cd0>,
<networkx.classes.digraph.DiGraph at 0x197cf5790>,
<networkx.classes.digraph.DiGraph at 0x186542490>,
<networkx.classes.digraph.DiGraph at 0x1988f6e90>,
<networkx.classes.digraph.DiGraph at 0x1988f67d0>,
<networkx.classes.digraph.DiGraph at 0x1951f2050>,
<networkx.classes.digraph.DiGraph at 0x198feeed0>,
<networkx.classes.digraph.DiGraph at 0x197e85650>,
<networkx.classes.digraph.DiGraph at 0x1914f8490>,
<networkx.classes.digraph.DiGraph at 0x19988c750>,
<networkx.classes.digraph.DiGraph at 0x1998806d0>,
<networkx.classes.digraph.DiGraph at 0x1999bd010>,
<networkx.classes.digraph.DiGraph at 0x199baa550>,
<networkx.classes.digraph.DiGraph at 0x199d9b690>,
<networkx.classes.digraph.DiGraph at 0x199e69550>,
<networkx.classes.digraph.DiGraph at 0x19a058ed0>,

```

```

<networkx.classes.digraph.DiGraph at 0x199e686d0>,
<networkx.classes.digraph.DiGraph at 0x19a57e0d0>,
<networkx.classes.digraph.DiGraph at 0x1996c5050>,
<networkx.classes.digraph.DiGraph at 0x1997cddb0>,
<networkx.classes.digraph.DiGraph at 0x18f897650>,
<networkx.classes.digraph.DiGraph at 0x19abb71d0>,
<networkx.classes.digraph.DiGraph at 0x19ad9dc10>,
<networkx.classes.digraph.DiGraph at 0x19b135090>],
'assortativity_means': {'In-In Assortativity': 0.3682335431525876,
'In-Out Assortativity': -0.3475729851689017,
'Out-In Assortativity': 0.12351272148538885,
'Out-Out Assortativity': -0.09066861341566969},
'assortativity_stds': {'In-In Assortativity': 0.0031598717755630425,
'In-Out Assortativity': 0.0023173275831781105,
'Out-In Assortativity': 0.004190586592479531,
'Out-Out Assortativity': 0.0051371284311176775},
'assortativity_data': {'In-In Assortativity': -0.06360116568177393,
'In-Out Assortativity': -0.06507198196482604,
'Out-In Assortativity': -0.06546711179930208,
'Out-Out Assortativity': -0.06730720421208614}},
'WDN_2001': {'nentropy_graph': <NEMtropy.graph_classes.UndirectedGraph at
0x143733850>,
'fit': None,
'error_degree': 8.608907364759943e-09,
'error_strength': None,
'samples': [<networkx.classes.digraph.DiGraph at 0x197e9cb90>,
<networkx.classes.digraph.DiGraph at 0x19b70be90>,
<networkx.classes.digraph.DiGraph at 0x19b6f51d0>,
<networkx.classes.digraph.DiGraph at 0x19b778250>,
<networkx.classes.digraph.DiGraph at 0x19bb3f050>,
<networkx.classes.digraph.DiGraph at 0x19bd26cd0>,
<networkx.classes.digraph.DiGraph at 0x19bdb0f90>,
<networkx.classes.digraph.DiGraph at 0x19c129090>,
<networkx.classes.digraph.DiGraph at 0x19c15e6d0>,
<networkx.classes.digraph.DiGraph at 0x19b720d90>,
<networkx.classes.digraph.DiGraph at 0x19c72f390>,
<networkx.classes.digraph.DiGraph at 0x19c95fd10>,
<networkx.classes.digraph.DiGraph at 0x19cb5d610>,
<networkx.classes.digraph.DiGraph at 0x19cd41f90>,
<networkx.classes.digraph.DiGraph at 0x19cf47f90>,
<networkx.classes.digraph.DiGraph at 0x19d155910>,
<networkx.classes.digraph.DiGraph at 0x19b7a97d0>,
<networkx.classes.digraph.DiGraph at 0x1982e6590>,
<networkx.classes.digraph.DiGraph at 0x19d963f90>,
<networkx.classes.digraph.DiGraph at 0x19d9997d0>,
<networkx.classes.digraph.DiGraph at 0x19d999b90>,
<networkx.classes.digraph.DiGraph at 0x19b720450>,

```

```

<networkx.classes.digraph.DiGraph at 0x19b7aa4d0>,
<networkx.classes.digraph.DiGraph at 0x19e0246d0>,
<networkx.classes.digraph.DiGraph at 0x19b661710>,
<networkx.classes.digraph.DiGraph at 0x19e3d9a10>,
<networkx.classes.digraph.DiGraph at 0x19e78afd0>,
<networkx.classes.digraph.DiGraph at 0x19e83c610>,
<networkx.classes.digraph.DiGraph at 0x19edb40d0>,
<networkx.classes.digraph.DiGraph at 0x19edb4dd0>],
'assortativity_means': {'In-In Assortativity': 0.38876608095688187,
'In-Out Assortativity': -0.3508887051454812,
'Out-In Assortativity': 0.14148898340696175,
'Out-Out Assortativity': -0.07422270202355051},
'assortativity_std': {'In-In Assortativity': 0.0029984214818470994,
'In-Out Assortativity': 0.0020843107279951345,
'Out-In Assortativity': 0.004569380141079644,
'Out-Out Assortativity': 0.004186917126078424},
'assortativity_data': {'In-In Assortativity': -0.060016931181439886,
'In-Out Assortativity': -0.061178239732267325,
'Out-In Assortativity': -0.06192625961852602,
'Out-Out Assortativity': -0.06348734473716487}}}

```

2.4 4. Plot strength assortativity as a function of time (the WTW network year), comparing the observed real value with the average and error bars obtained from the samples samples.

Hint: plot error bars as confidence intervals. Plot all pairs of assortativity for the undirected and directed case(in-in, in-out, out-out)

```
[187]: assortativities
```

```

[187]: {'WDN_1997': {'In-In Assortativity': -0.07368635207387701,
'In-Out Assortativity': -0.07132955497657792,
'Out-In Assortativity': -0.07171683897072376,
'Out-Out Assortativity': -0.06970079872009399},
'WDN_1992': {'In-In Assortativity': -0.04868776409103533,
'In-Out Assortativity': -0.046647404340923876,
'Out-In Assortativity': -0.059869793086991335,
'Out-Out Assortativity': -0.05760318920541264},
'WDN_2000': {'In-In Assortativity': -0.0631970717497539,
'In-Out Assortativity': -0.06467170213846346,
'Out-In Assortativity': -0.06539073425903164,
'Out-Out Assortativity': -0.06730072607061888},
'WDN_1998': {'In-In Assortativity': -0.0655705781839572,
'In-Out Assortativity': -0.06540616504184192,
'Out-In Assortativity': -0.06572057397949527,
'Out-Out Assortativity': -0.06583617855060356},
'WDN_1994': {'In-In Assortativity': -0.07868988970611883,

```

```

'In-Out Assortativity': -0.07622523439269807,
'Out-In Assortativity': -0.08022501617948047,
'Out-Out Assortativity': -0.07800143058821393},
'WDN_1993': {'In-In Assortativity': -0.06537929984057925,
'In-Out Assortativity': -0.062189867187614475,
'Out-In Assortativity': -0.07202068872957715,
'Out-Out Assortativity': -0.06910618411247677},
'WDN_2002': {'In-In Assortativity': -0.06019317176336694,
'In-Out Assortativity': -0.060795564663496796,
'Out-In Assortativity': -0.06303814946654555,
'Out-Out Assortativity': -0.06417872159686847},
'WDN_1996': {'In-In Assortativity': -0.07686964594098099,
'In-Out Assortativity': -0.07546909805023332,
'Out-In Assortativity': -0.07535502512416227,
'Out-Out Assortativity': -0.07420828746119328},
'WDN_1995': {'In-In Assortativity': -0.07921393553292572,
'In-Out Assortativity': -0.07702100817504068,
'Out-In Assortativity': -0.07830645801950578,
'Out-Out Assortativity': -0.07638722326404519},
'WDN_1999': {'In-In Assortativity': -0.06360116568177393,
'In-Out Assortativity': -0.06507198196482604,
'Out-In Assortativity': -0.06546711179930208,
'Out-Out Assortativity': -0.06730720421208614},
'WDN_2001': {'In-In Assortativity': -0.060016931181439886,
'In-Out Assortativity': -0.061178239732267325,
'Out-In Assortativity': -0.06192625961852602,
'Out-Out Assortativity': -0.06348734473716487}}

```

```

[188]: # Assortativity pairs
assortativity_pairs = ['In-In Assortativity', 'In-Out Assortativity', 'Out-In_
↳Assortativity', 'Out-Out Assortativity']

```

```

[189]: # Years
years = list(nemtrophy_directed_results.keys())
x = np.arange(len(years))

# Plotting
plt.figure(figsize=(18, 6))

for i, pair in enumerate(assortativity_pairs):
    plt.subplot(1, len(assortativity_pairs), i + 1)
    sample_means =
↳[nemtrophy_directed_results[year]['assortativity_means'][pair] for year in
↳years]
    sample_std = [nemtrophy_directed_results[year]['assortativity_stds'][pair]
↳for year in years]

```

```

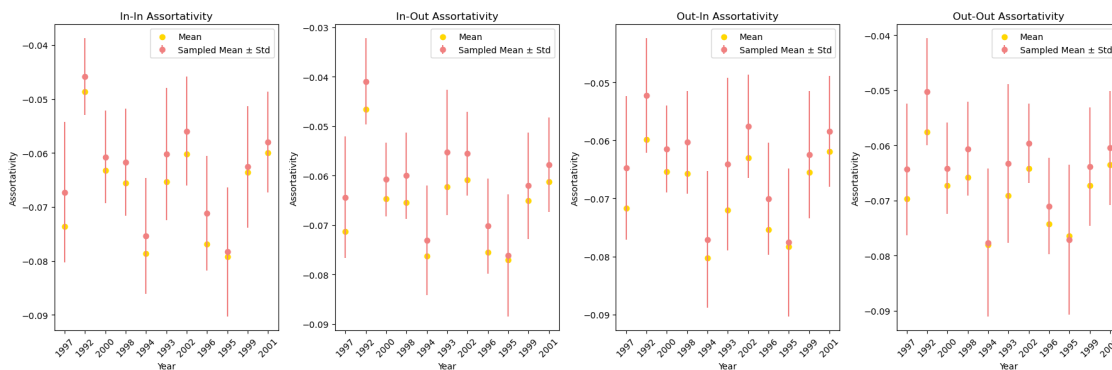
plt.errorbar(x, sample_means, yerr=sample_std, fmt='o', label='Sampled Mean',
↳ ± Std', color=COLOR_RED)

real_mean = [nentropy_directed_results[year]['assortativity_data'][pair]
↳ for year in years]
plt.scatter(x, real_mean, color=COLOR_YELLOW, label='Mean')

plt.xticks(x, [year[-4:] for year in years], rotation=45)
plt.title(f'{pair}')
plt.xlabel('Year')
plt.ylabel('Assortativity')
plt.legend()

plt.tight_layout()
plt.show()

```



```

[190]: # Years
years = list(nentropy_undirected_results.keys())
x = np.arange(len(years))

# Plotting
plt.figure(figsize=(18, 6))

for i, pair in enumerate(assortativity_pairs):
    plt.subplot(1, len(assortativity_pairs), i + 1)
    sample_means =
↳ [nentropy_undirected_results[year]['assortativity_means'][pair] for year in
↳ years]
    sample_std = [nentropy_undirected_results[year]['assortativity_stds'][pair]
↳ for year in years]

```

```

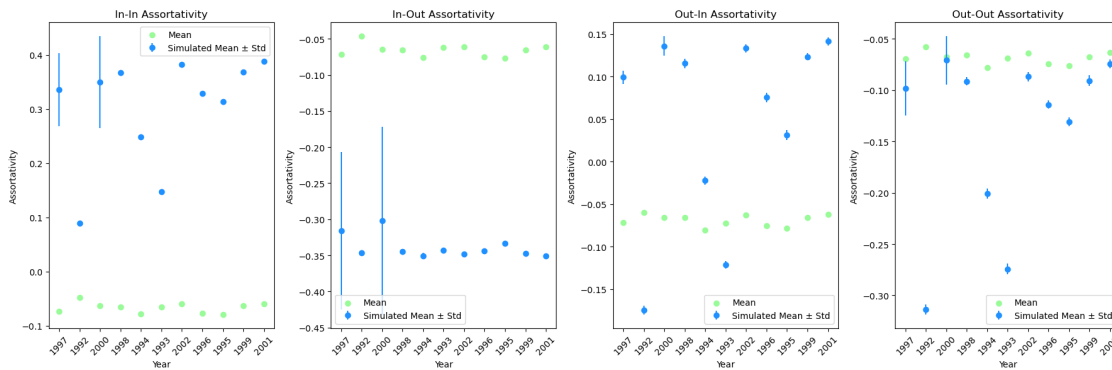
plt.errorbar(x, sample_means, yerr=sample_std, fmt='o', label='Simulated_
↳Mean ± Std', color=COLOR_BLUE)

# TODO: Can we use the results from the original directed data?
real_mean = [nentropy_directed_results[year]['assortativity_data'][pair]_
↳for year in years]
plt.scatter(x, real_mean, color=COLOR_GREEN, label='Mean')

plt.xticks(x, [year[-4:] for year in years], rotation=45)
plt.title(f'{pair}')
plt.xlabel('Year')
plt.ylabel('Assortativity')
plt.legend()

plt.tight_layout()
plt.show()

```



2.5 5. Write a short paragraph to comment on the interpretation of strength assortativity for this dataset, and about the conclusions you can draw via the inference you made using the ERGM models.

Creating Results: jupyter nbconvert Assignment_03.ipynb --to pdf --output Assignment_03_Results.pdf